# 종합실습 4 - E-Commerce 데이터 요청 및 SQL 튜닝 리포트

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DBMS: PostgreSQL 17.10 (Homebrew)

실행환경: Local PC / DBeaver Community

측정 명령: EXPLAIN (ANALYZE, BUFFERS, SUMMARY)

> 시드 데이터는 random()으로 생성되므로 실행 결과와 시간은 환경마다 달라질 수 있다. 절대시간뿐 아니라 Scan/Join 방식, 반복 횟수, 예상·실제 행 수와 Buffers를 함께 비교한다.

## 1. 실습 목적

구매·재무 등 다른 실무 팀에서 요청한 E-Commerce 자료를 정확한 기준으로 산출하고, 같은 결과를 더 효율적으로 반환하도록 SQL과 인덱스를 개선한다. Q1~Q10은 튜닝 전후 쿼리와 실행계획을 비교하고, PostgreSQL의 세 가지 물리 조인과 Materialized View 활용 방법을 검토한다.

## 2. 공통 데이터 기준

| 항목 | 정의 |
|---|---|
| 실매출 주문 | order_status IN ('paid', 'shipped', 'delivered') |
| 주문 매출 | SUM(order_items.line_total) |
| AOV | 실매출 합계 ÷ 실매출 주문 수 |
| Q1 기간 | 실행시점 기준 최근 1개월 |
| Q3 기간 | 실행시점 기준 최근 90일 |
| Q6 재구매 | 첫 구매 후 30일 이내 추가 실구매 |
| Q10 대상 | 전체 3,000명 누적매출 상위 1%인 30명 |
| Q10 표시금액 | 선정된 30명의 최근 60일 실매출 |

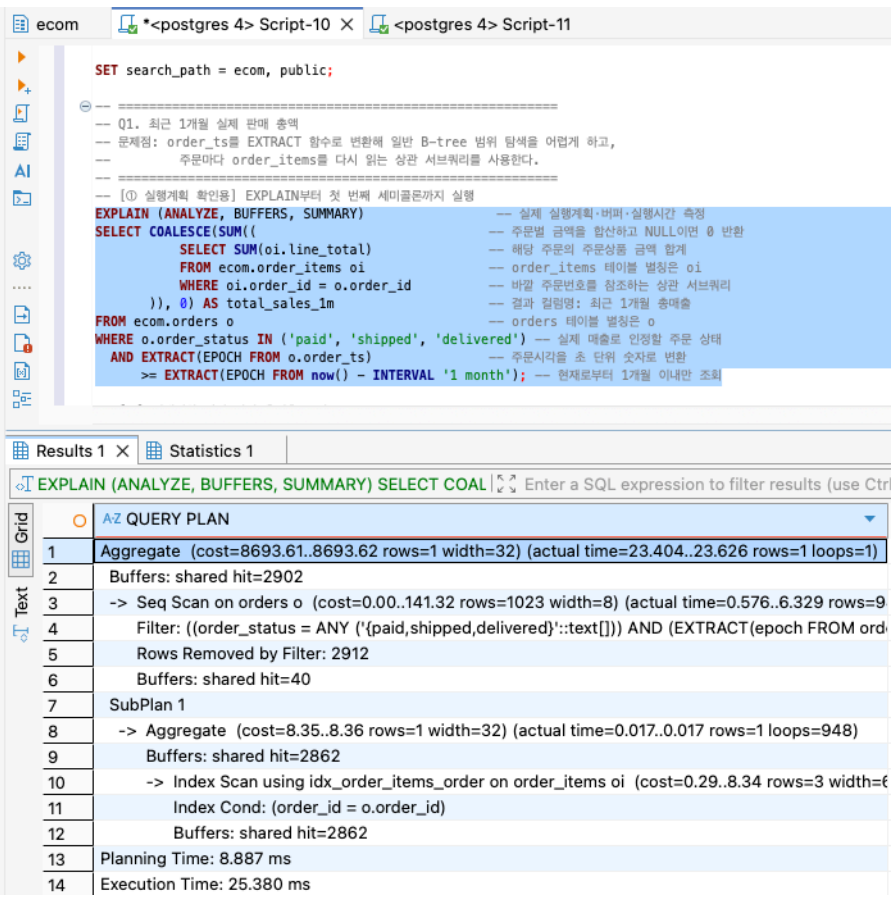
## 3. Q1 - 최근 1개월 실제 판매 총액

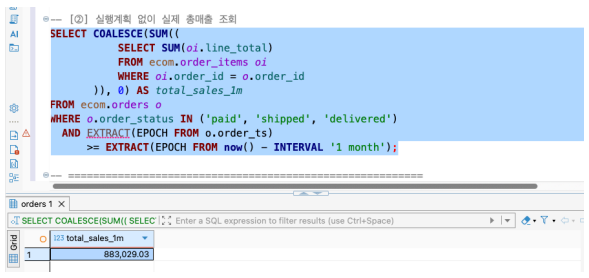
### 요청사항

paid, shipped, delivered 상태이면서 now() - interval '1 month' 이후인 주문의 line_total 합계를 제공한다.

### 튜닝 전 쿼리·결과 화면

Q1 튜닝 전 - 다음 페이지




## 느린 이유

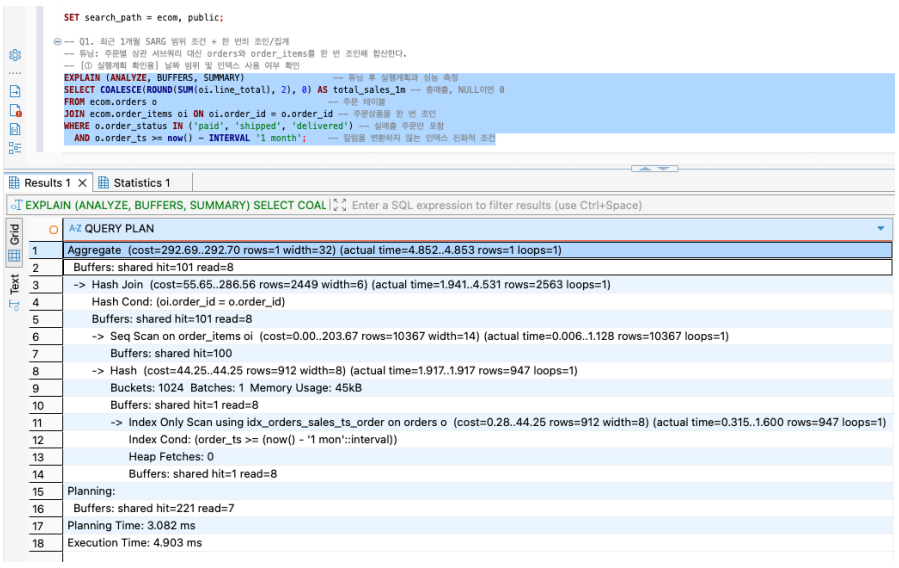
`order_ts`를 함수로 변환하면 일반 날짜 인덱스의 범위탐색이 어려워질 수 있다. 또한 주문마다 주문상품 합계를 상관 서브쿼리로 반복 조회한다.

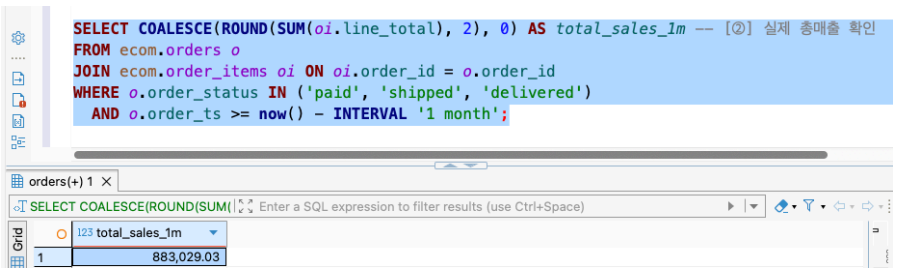
## 개선 방법

날짜 컬럼을 그대로 둔 SARG 범위 조건을 사용하고, 주문과 주문상품을 한 번 조인해 전체 합계를 계산한다. 실매출 상태만 포함하는 부분 인덱스로 대상 주문 탐색 범위를 줄인다.

## 튜닝 후 쿼리·결과 화면

Q1 튜닝 후




| 구분 | Scan/Join 방식 | Buffers | Execution Time | 결과 금액 |
|---|---|---|---|---|
| 튜닝 전 | Seq Scan + SubPlan | hit=2,902 | 25.380 ms | 883,029.03 |
| 튜닝 후 | Index Only + Hash | hit=101/read=8 | 4.903 ms | 883,029.03 |

**결과 해석:** 전후 결과는 동일하며 실행시간이 80.7% 단축되고 반복 SubPlan이 제거되었다.

# 4. Q2 - 월별 주문 수·매출·AOV

## 요청사항

월별 실매출 주문 수, 총매출, 주문당 평균 금액을 제공한다.

Q2 튜닝 전

```sql
-- ============================================================
-- Q2. 월별 주문 수, 매출, AOV
-- 문제점: 상세 행을 먼저 모두 조인한 뒤 COUNT(DISTINCT)로 중복을 제거한다.
-- ============================================================
-- [① 실행계획 확인용] 월별 집계의 Scan/Join과 실행시간 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)                    -- 실제 실행계획과 성능 측정
SELECT date_trunc('month', o.order_ts)::date AS sales_month, -- 주문시각을 월 단위로 묶음
       COUNT(DISTINCT o.order_id) AS order_count,  -- 상세행 중복을 제거한 주문 수
       ROUND(SUM(oi.line_total), 2) AS revenue,    -- 월별 매출 합계, 소수점 둘째 자리
       ROUND(SUM(oi.line_total) / NULLIF(COUNT(DISTINCT o.order_id), 0), 2) AS aov --
FROM ecom.orders o                                 -- 주문 테이블
JOIN ecom.order_items oi ON oi.order_id = o.order_id -- 주문번호로 주문상품 연결
WHERE o.order_status IN ('paid', 'shipped', 'delivered') -- 실매출 주문만 포함
GROUP BY 1                                         -- SELECT 첫 번째 컬럼인 월로 그룹화
ORDER BY 1;                                        -- 월 오름차순 정렬
```

Results 1 | Statistics 1

EXPLAIN (ANALYZE, BUFFERS, SU | Enter a SQL expression to filter results (use Ctrl+Space)

```
QUERY PLAN
GroupAggregate  (cost=939.49..1113.93 rows=3068 width=76) (actual time=10.469..11.600 rows=6 loops=1)
  Group Key: ((date_trunc('month'::text, o.order_ts))::date)
  Buffers: shared hit=140
  -> Sort  (cost=939.49..960.09 rows=8240 width=18) (actual time=10.362..10.640 rows=8230 loops=1)
     Sort Key: ((date_trunc('month'::text, o.order_ts))::date), o.order_id
     Sort Method: quicksort  Memory: 706kB
     Buffers: shared hit=140
     -> Hash Join  (cost=131.43..403.54 rows=8240 width=18) (actual time=1.613..7.866 rows=8230 loops=1)
        Hash Cond: (oi.order_id = o.order_id)
        Buffers: shared hit=140
        -> Seq Scan on order_items oi  (cost=0.00..203.67 rows=10367 width=14) (actual time=0.009..1.122 rows=10367 loops=1)
           Buffers: shared hit=100
        -> Hash  (cost=93.08..93.08 rows=3068 width=16) (actual time=1.543..1.543 rows=3068 loops=1)
           Buckets: 4096  Batches: 1  Memory Usage: 176kB
           Buffers: shared hit=40
           -> Seq Scan on orders o  (cost=0.00..93.08 rows=3068 width=16) (actual time=0.016..1.071 rows=3068 loops=1)
              Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
              Rows Removed by Filter: 792
              Buffers: shared hit=40
Planning:
  Buffers: shared hit=49 dirtied=1
Planning Time: 1.979 ms
Execution Time: 11.646 ms
```

```sql
SELECT date_trunc('month', o.order_ts)::date AS sales_month, -- [②] 실제 월별 결과 조회
       COUNT(DISTINCT o.order_id) AS order_count,
       ROUND(SUM(oi.line_total), 2) AS revenue,
       ROUND(SUM(oi.line_total) / NULLIF(COUNT(DISTINCT o.order_id), 0), 2) AS aov
FROM ecom.orders o
JOIN ecom.order_items oi ON oi.order_id = o.order_id
WHERE o.order_status IN ('paid', 'shipped', 'delivered')
GROUP BY 1
ORDER BY 1;
-- ============================================================
```

orders(+) 1

SELECT date_trunc('month', o.ord | Enter a SQL expression to filter results (use Ctrl+Space)

| | sales_month | order_count | revenue | aov |
|---|---|---|---|---|
| 1 | 2026-04-01 | 267 | 228,521.46 | 855.89 |
| 2 | 2026-05-01 | 673 | 551,927.62 | 820.1 |
| 3 | 2026-06-01 | 794 | 717,531.35 | 903.69 |
| 4 | 2026-07-01 | 892 | 829,725.16 | 930.19 |
| 5 | 2026-08-01 | 439 | 390,509.57 | 889.54 |
| 6 | 2026-09-01 | 3 | 812.51 | 270.84 |

## 느린 이유와 개선 방법

상세 주문상품을 먼저 조인한 상태에서는 주문이 여러 행으로 증가하므로 `COUNT(DISTINCT order_id)`가 필요하다. 튜닝 후에는 주문별 금액을 먼저 집계한 다음 월별로 다시 집계해 중복 제거 비용을 줄인다.

Q2 튜닝 후

```sql
-- Q2. 주문별 매출을 먼저 계산해 COUNT(DISTINCT)를 제거
-- 튜닝: 주문별로 한 행을 만든 뒤 중복 제거 없이 order_id를 집계한다.
-- [① 실행계획 확인용] 주문별 선집계 후 월별 집계 계획 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)
WITH order_totals AS (                          -- 주문별 금액을 먼저 만드는 CTE
    SELECT o.order_id,
           date_trunc('month', o.order_ts)::date AS sales_month,
           SUM(oi.line_total) AS order_total       -- 주문상품을 주문 단위로 합산
    FROM ecom.orders o
    JOIN ecom.order_items oi ON oi.order_id = o.order_id
    WHERE o.order_status IN ('paid', 'shipped', 'delivered')
    GROUP BY o.order_id, o.order_ts
)
SELECT sales_month,
       COUNT(order_id) AS order_count,             -- 주문당 한 행이므로 주문번호 집계
       ROUND(SUM(order_total), 2) AS revenue,
       ROUND(AVG(order_total), 2) AS aov
FROM order_totals
GROUP BY sales_month
ORDER BY sales_month;
```

Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) WITH order_totals AS ( SELEC | Enter a SQL expression to filter results (use Ctrl+Space)

```
QUERY PLAN
Sort  (cost=522.57..523.07 rows=200 width=76) (actual time=6.733..6.735 rows=6 loops=1)
  Sort Key: ((date_trunc('month'::text, o.order_ts))::date)
  Sort Method: quicksort  Memory: 25kB
  Buffers: shared hit=143
  -> HashAggregate  (cost=510.92..514.92 rows=200 width=76) (actual time=6.714..6.721 rows=6 loops=1)
     Group Key: (date_trunc('month'::text, o.order_ts))::date
     Batches: 1  Memory Usage: 40kB
     Buffers: shared hit=140
     -> HashAggregate  (cost=403.54..457.23 rows=3068 width=52) (actual time=5.123..6.324 rows=3068 loops=1)
        Group Key: o.order_id
        Batches: 1  Memory Usage: 1393kB
        Buffers: shared hit=140
        -> Hash Join  (cost=131.43..362.34 rows=8240 width=22) (actual time=0.906..3.375 rows=8230 loops=1)
           Hash Cond: (oi.order_id = o.order_id)
           Buffers: shared hit=140
           -> Seq Scan on order_items oi  (cost=0.00..203.67 rows=10367 width=14) (actual time=0.010..0.824 rows=10367 loops=1)
              Buffers: shared hit=100
           -> Hash  (cost=93.08..93.08 rows=3068 width=16) (actual time=0.879..0.879 rows=3068 loops=1)
              Buckets: 4096  Batches: 1  Memory Usage: 176kB
              Buffers: shared hit=40
              -> Seq Scan on orders o  (cost=0.00..93.08 rows=3068 width=16) (actual time=0.007..0.603 rows=3068 loops=1)
                 Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                 Rows Removed by Filter: 792
                 Buffers: shared hit=40
Planning:
  Buffers: shared hit=69 read=7
Planning Time: 5.743 ms
Execution Time: 6.879 ms
```

```sql
WITH order_totals AS (                          -- [②] 실제 월별 결과 조회
    SELECT o.order_id, date_trunc('month', o.order_ts)::date AS sales_month,
           SUM(oi.line_total) AS order_total
    FROM ecom.orders o
    JOIN ecom.order_items oi ON oi.order_id = o.order_id
    WHERE o.order_status IN ('paid', 'shipped', 'delivered')
    GROUP BY o.order_id, o.order_ts
)
SELECT sales_month, COUNT(order_id) AS order_count,
       ROUND(SUM(order_total), 2) AS revenue,
       ROUND(AVG(order_total), 2) AS aov
FROM order_totals
GROUP BY sales_month
ORDER BY sales_month;
```

Results 1

WITH order_totals AS ( SELECT o.order_id, date_trunc('month', o.order_ts | Enter a SQL expression to filter res

| | sales_month | order_count | revenue | aov |
|---|---|---|---|---|
| 1 | 2026-04-01 | 267 | 228,521.46 | 855.89 |
| 2 | 2026-05-01 | 673 | 551,927.62 | 820.1 |
| 3 | 2026-06-01 | 794 | 717,531.35 | 903.69 |
| 4 | 2026-07-01 | 892 | 829,725.16 | 930.19 |
| 5 | 2026-08-01 | 439 | 390,509.57 | 889.54 |
| 6 | 2026-09-01 | 3 | 812.51 | 270.84 |

| 구분 | 집계 방식 | Buffers | Execution Time | 전후 결과 동일 여부 |
|---|---|---|---|---|
| 튜닝 전 | 상세행 + `COUNT(DISTINCT)` | hit=140 | 11.646 ms | - |
| 튜닝 후 | 주문별 선집계 + `COUNT(order_id)` | hit=143 | 6.879 ms | 동일 |

**결과 해석:** 주문별 선집계로 DISTINCT 제거 비용을 줄여 실행시간이 40.9% 단축되었다.

# 5. Q3 - 최근 90일 카테고리 매출 Top 10

## 요청사항

최근 90일 실매출을 카테고리별로 집계해 상위 10개를 제공한다.

Q3 튜닝 전

```sql
-- ==========================================================
-- Q3. 최근 90일 카테고리 Top 10
-- 문제점: 비-SARG 날짜 조건을 사용하고 상세 데이터를 그대로 집계한다.
-- ==========================================================
-- [① 실행계획 확인용] 최근 90일 필터와 테이블 조인 계획 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)                  -- 실제 실행계획과 성능 측정
SELECT c.category_id,
       c.category_name,
       ROUND(SUM(oi.line_total), 2) AS revenue_90d -- 카테고리별 최근 90일 매출
FROM ecom.orders o                                   -- 주문에서 시작
JOIN ecom.order_items oi ON oi.order_id = o.order_id -- 주문상품 연결
JOIN ecom.products p ON p.product_id = oi.product_id -- 상품 연결
JOIN ecom.categories c ON c.category_id = p.category_id -- 카테고리 연결
WHERE o.order_status IN ('paid', 'shipped', 'delivered') -- 실매출 주문만 포함
  AND o.order_ts::date >= CURRENT_DATE - 90          -- timestamp를 date로 변환해 최근 90일 필터
GROUP BY c.category_id, c.category_name              -- 카테고리별로 묶어 합산
ORDER BY revenue_90d DESC, c.category_id             -- 매출이 큰 순서로 정렬
LIMIT 10;                                            -- 상위 10개만 표시
```

Results 1 | Statistics 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) SELECT c.cate

| | QUERY PLAN |
|---|---|
| 1 | Limit (cost=497.38..497.41 rows=10 width=72) (actual time=11.220..11.225 rows=10 loops=1) |
| 2 | Buffers: shared hit=148 |
| 3 | -> Sort (cost=497.38..500.06 rows=1070 width=72) (actual time=11.219..11.222 rows=10 loops=1) |
| 4 | Sort Key: (round(sum(oi.line_total), 2)) DESC, c.category_id |
| 5 | Sort Method: quicksort Memory: 25kB |
| 6 | Buffers: shared hit=148 |
| 7 | -> HashAggregate (cost=458.21..474.26 rows=1070 width=72) (actual time=10.966..10.979 rows=10 loops=1) |
| 8 | Group Key: c.category_id |
| 9 | Batches: 1 Memory Usage: 73kB |
| 10 | Buffers: shared hit=148 |
| 11 | -> Hash Join (cost=199.04..444.47 rows=2748 width=46) (actual time=3.828..9.543 rows=6662 loops=1) |
| 12 | Hash Cond: (p.category_id = c.category_id) |
| 13 | Buffers: shared hit=148 |
| 14 | -> Hash Join (cost=164.96..403.15 rows=2748 width=14) (actual time=3.311..7.831 rows=6662 loops=1) |
| 15 | Hash Cond: (oi.product_id = p.product_id) |
| 16 | Buffers: shared hit=147 |
| 17 | -> Hash Join (cost=144.46..375.38 rows=2748 width=14) (actual time=2.841..6.120 rows=6662 loops=1) |
| 18 | Hash Cond: (oi.order_id = o.order_id) |
| 19 | Buffers: shared hit=140 |
| 20 | -> Seq Scan on order_items oi (cost=0.00..203.67 rows=10367 width=22) (actual time=0.015..1.308 rows=10367 loops=1) |
| 21 | Buffers: shared hit=100 |
| 22 | -> Hash (cost=131.68..131.68 rows=1023 width=8) (actual time=2.502..2.502 rows=2483 loops=1) |
| 23 | Buckets: 4096 (originally 1024) Batches: 1 (originally 1) Memory Usage: 129kB |
| 24 | Buffers: shared hit=40 |

Results 1 | Statistics 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) SELECT c.cate

| | QUERY PLAN |
|---|---|
| 20 | -> Seq Scan on order_items oi (cost=0.00..203.67 rows=10367 width=22) (actual time=0.015..1.308 rows=10367 loops=1) |
| 21 | Buffers: shared hit=100 |
| 22 | -> Hash (cost=131.68..131.68 rows=1023 width=8) (actual time=2.502..2.502 rows=2483 loops=1) |
| 23 | Buckets: 4096 (originally 1024) Batches: 1 (originally 1) Memory Usage: 129kB |
| 24 | Buffers: shared hit=40 |
| 25 | -> Seq Scan on orders o (cost=0.00..131.68 rows=1023 width=8) (actual time=0.023..1.672 rows=2483 loops=1) |
| 26 | Filter: ((order_status = ANY ('{paid,shipped,delivered}'::text[])) AND ((order_ts)::date >= (CURRENT_DATE - 90))) |
| 27 | Rows Removed by Filter: 1377 |
| 28 | Buffers: shared hit=40 |
| 29 | -> Hash (cost=13.00..13.00 rows=600 width=16) (actual time=0.449..0.450 rows=600 loops=1) |
| 30 | Buckets: 1024 Batches: 1 Memory Usage: 37kB |
| 31 | Buffers: shared hit=7 |
| 32 | -> Seq Scan on products p (cost=0.00..13.00 rows=600 width=16) (actual time=0.006..0.368 rows=600 loops=1) |
| 33 | Buffers: shared hit=7 |
| 34 | -> Hash (cost=20.70..20.70 rows=1070 width=40) (actual time=0.276..0.276 rows=14 loops=1) |
| 35 | Buckets: 2048 Batches: 1 Memory Usage: 17kB |
| 36 | Buffers: shared hit=1 |
| 37 | -> Seq Scan on categories c (cost=0.00..20.70 rows=1070 width=40) (actual time=0.256..0.257 rows=14 loops=1) |
| 38 | Buffers: shared hit=1 |
| 39 | Planning: |
| 40 | Buffers: shared hit=27 |
| 41 | Planning Time: 6.885 ms |
| 42 | Execution Time: 11.863 ms |

```sql
SELECT c.category_id,                                -- [②] 실제 Top 10 결과 조회
       c.category_name,
       ROUND(SUM(oi.line_total), 2) AS revenue_90d
FROM ecom.orders o
JOIN ecom.order_items oi ON oi.order_id = o.order_id
JOIN ecom.products p ON p.product_id = oi.product_id
JOIN ecom.categories c ON c.category_id = p.category_id
WHERE o.order_status IN ('paid', 'shipped', 'delivered')
  AND o.order_ts::date >= CURRENT_DATE - 90
GROUP BY c.category_id, c.category_name
ORDER BY revenue_90d DESC, c.category_id
LIMIT 10;

-- ==========================================================
-- 04. 제품별 누적매출 RANK() Top 20
```

categories 1

SELECT c.category_id, c.category_name, ROUND(SUM(oi.

| | category_id | category_name | revenue_90d |
|---|---|---|---|
| 1 | 13 | Outdoor | 758,672.65 |
| 2 | 2 | Phones | 541,480.74 |
| 3 | 6 | Appliances | 138,788.46 |
| 4 | 14 | Fitness | 125,711.82 |
| 5 | 11 | Shoes | 124,829.38 |
| 6 | 7 | Cookware | 117,640.32 |
| 7 | 4 | Audio | 112,653.77 |
| 8 | 3 | Laptops | 109,500.6 |
| 9 | 9 | Men | 99,046.69 |
| 10 | 10 | Women | 96,304.36 |

## 느린 이유와 개선 방법

비-SARG 날짜 조건은 인덱스 범위 탐색을 방해할 수 있다. 튜닝 후에는 최근 실매출 주문을 먼저 축소하고 `order_items → products → categories` 순서로 연결한다.

Q3 튜닝 후

```sql
-- Q3. 기간/상태로 orders를 먼저 축소하고 SARG 범위 조건 사용
-- 튜닝: 최근 90일 실매출 주문만 먼저 골라 조인해야 할 행 수를 줄인다.
-- [① 실행계획 확인용] 최근 주문 선필터와 조인 계획 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)
WITH recent_sales AS MATERIALIZED (                  -- 최근 주문을 먼저 계산해 저장하는 CTE
    SELECT order_id
    FROM ecom.orders
    WHERE order_status IN ('paid', 'shipped', 'delivered')
      AND order_ts >= CURRENT_DATE - 90              -- 컬럼을 변환하지 않는 최근 90일 조건
)
SELECT c.category_id,
       c.category_name,
       ROUND(SUM(oi.line_total), 2) AS revenue_90d
FROM recent_sales rs
JOIN ecom.order_items oi ON oi.order_id = rs.order_id
JOIN ecom.products p ON p.product_id = oi.product_id
JOIN ecom.categories c ON c.category_id = p.category_id
GROUP BY c.category_id, c.category_name
ORDER BY revenue_90d DESC, c.category_id
LIMIT 10;
```

```sql
WITH recent_sales AS MATERIALIZED (                  -- [②] 실제 카테고리 Top 10 조회
    SELECT order_id
    FROM ecom.orders
    WHERE order_status IN ('paid', 'shipped', 'delivered')
      AND order_ts >= CURRENT_DATE - 90
)
SELECT c.category_id, c.category_name,
       ROUND(SUM(oi.line_total), 2) AS revenue_90d
FROM recent_sales rs
JOIN ecom.order_items oi ON oi.order_id = rs.order_id
JOIN ecom.products p ON p.product_id = oi.product_id
JOIN ecom.categories c ON c.category_id = p.category_id
GROUP BY c.category_id, c.category_name
ORDER BY revenue_90d DESC, c.category_id
LIMIT 10;
```

categories 1

WITH recent_sales AS MATERIALIZED ( SELECT order_id I

| | category_id | category_name | revenue_90d |
|---|---|---|---|
| 1 | 13 | Outdoor | 758,672.65 |
| 2 | 2 | Phones | 541,480.74 |
| 3 | 6 | Appliances | 138,788.46 |
| 4 | 14 | Fitness | 125,711.82 |
| 5 | 11 | Shoes | 124,829.38 |
| 6 | 7 | Cookware | 117,640.32 |
| 7 | 4 | Audio | 112,653.77 |
| 8 | 3 | Laptops | 109,500.6 |
| 9 | 9 | Men | 99,046.69 |
| 10 | 10 | Women | 96,304.36 |

Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) WITH recent_

```
Limit  (cost=659.69..659.71 rows=10 width=72) (actual time=11.383..11.387 rows=10 loops=1)
  Buffers: shared hit=120 read=9
  CTE recent_sales
    -> Index Only Scan using idx_orders_sales_ts_order on orders  (cost=0.28..111.02 rows=2442 width=8) (actual time=0.413..1.477 rows=2483 loops=1)
        Index Cond: (order_ts >= (CURRENT_DATE - 90))
        Heap Fetches: 0
        Buffers: shared hit=12 read=9
  -> Sort  (cost=548.67..551.34 rows=1070 width=72) (actual time=11.382..11.385 rows=10 loops=1)
        Sort Key: (round(sum(oi.line_total), 2)) DESC, c.category_id
        Sort Method: quicksort  Memory: 25kB
        Buffers: shared hit=120 read=9
        -> HashAggregate  (cost=509.50..525.55 rows=1070 width=72) (actual time=11.354..11.369 rows=10 loops=1)
            Group Key: c.category_id
            Batches: 1  Memory Usage: 73kB
            Buffers: shared hit=120 read=9
            -> Hash Join  (cost=133.94..476.70 rows=6559 width=46) (actual time=2.623..9.671 rows=6662 loops=1)
                Hash Cond: (p.category_id = c.category_id)
                Buffers: shared hit=120 read=9
                -> Hash Join  (cost=99.87..425.34 rows=6559 width=14) (actual time=2.587..8.147 rows=6662 loops=1)
                    Hash Cond: (oi.product_id = p.product_id)
                    Buffers: shared hit=119 read=9
                    -> Hash Join  (cost=79.37..387.50 rows=6559 width=14) (actual time=2.372..6.349 rows=6662 loops=1)
                        Hash Cond: (oi.order_id = rs.order_id)
                        Buffers: shared hit=112 read=9
                        -> Seq Scan on order_items oi  (cost=0.00..203.67 rows=10367 width=22) (actual time=0.016..1.415 rows=10367 loops=1)
                            Buffers: shared hit=100
                        -> Hash  (cost=48.84..48.84 rows=2442 width=8) (actual time=2.335..2.335 rows=2483 loops=1)
                            Buffers: shared hit=12 read=9
                            -> CTE Scan on recent_sales rs  (cost=0.00..48.84 rows=2442 width=8) (actual time=0.414..2.010 rows=2483 loops=1)
                                Buffers: shared hit=12 read=9
                    -> Hash  (cost=13.00..13.00 rows=600 width=16) (actual time=0.194..0.194 rows=600 loops=1)
                        Buffers: shared hit=7
                        -> Seq Scan on products p  (cost=0.00..13.00 rows=600 width=16) (actual time=0.008..0.101 rows=600 loops=1)
                            Buffers: shared hit=7
                -> Hash  (cost=20.70..20.70 rows=1070 width=40) (actual time=0.019..0.019 rows=14 loops=1)
                    Buffers: shared hit=1
                    -> Seq Scan on categories c  (cost=0.00..20.70 rows=1070 width=40) (actual time=0.008..0.010 rows=14 loops=1)
                        Buffers: shared hit=1
Planning:
  Buffers: shared hit=64
Planning Time: 2.458 ms
Execution Time: 11.534 ms
```

| 구분 | Orders 접근 방식 | Join 방식 | Buffers | Execution Time |
|---|---|---|---|---|
| 튜닝 전 | Seq Scan | Hash Join | hit=148 | 11.863 ms |
| 튜닝 후 | Index Only Scan | Hash Join | hit=120/read=9 | 11.534 ms |

**Top 10 결과 및 해석:** Outdoor가 758,672.65로 1위였으며 결과는 동일하고 실행시간은 2.8% 단축되었다.

# 6. Q4 - 제품별 누적매출 RANK Top 20

## 요청사항

제품별 누적매출을 먼저 계산하고 `RANK()`로 상위 20개 제품을 제공한다.

Q4 튜닝 전

ecom | *<postgres 4> Script-10 | <postgres 4> Script-11

```
-- ==========================================================
-- Q4. 제품별 누적매출 RANK() Top 20
-- 문제점: 600개 제품 각각에 대해 매출 상관 서브쿼리를 실행한다.
-- ==========================================================
-- [① 실행계획 확인용] 반복 SubPlan의 loops와 실행시간 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)                 -- 실제 실행계획과 반복 횟수 측정
WITH product_revenue AS (                           -- 제품별 매출 중간 결과 CTE 시작
    SELECT p.product_id,
           p.product_name,
           COALESCE((                               -- 매출이 없으면 NULL 대신 0
               SELECT SUM(oi.line_total)            -- 제품별 주문상품 금액 합산
               FROM ecom.order_items oi
               JOIN ecom.orders o ON o.order_id = oi.order_id
               WHERE oi.product_id = p.product_id
                 AND o.order_status IN ('paid', 'shipped', 'delivered')
           ), 0) AS revenue
    FROM ecom.products p
), ranked AS (                                      -- 매출 결과에 순위를 붙이는 두 번째 CTE
    SELECT product_id,
           product_name,
           revenue,
           RANK() OVER (ORDER BY revenue DESC) AS revenue_rank -- 매출 내림차순 공동 순위
    FROM product_revenue
    WHERE revenue > 0
)
SELECT product_id, product_name, ROUND(revenue, 2) AS revenue, revenue_rank
FROM ranked
WHERE revenue_rank <= 20                            -- 순위가 20위 이내인 제품만 표시
ORDER BY revenue_rank, product_id;
```

Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) WITH product

```
Sort  (cost=158635.79..158636.29 rows=200 width=59) (actual time=36.685..36.687 rows=20 loops=1)
  Sort Key: ranked.revenue_rank, ranked.product_id
  Sort Method: quicksort  Memory: 26kB
  Buffers: shared hit=70677
  -> Subquery Scan on ranked  (cost=126903.76..158628.14 rows=200 width=59) (actual time=36.334..36.354 rows=20 loops=1)
      Buffers: shared hit=70677
      -> WindowAgg  (cost=126903.76..158627.64 rows=200 width=59) (actual time=36.332..36.349 rows=20 loops=1)
          Run Condition: (rank() OVER (?) <= 20)
          Buffers: shared hit=70677
          -> Sort  (cost=126903.74..126904.24 rows=200 width=51) (actual time=36.305..36.308 rows=21 loops=1)
              Sort Key: (COALESCE((SubPlan 1), '0'::numeric)) DESC
              Sort Method: quicksort  Memory: 52kB
              Buffers: shared hit=70677
              -> Seq Scan on products p  (cost=0.00..126896.10 rows=200 width=51) (actual time=1.089..35.299 rows=585 loops=1)
                  Filter: (COALESCE((SubPlan 2), '0'::numeric) > '0'::numeric)
                  Rows Removed by Filter: 15
```

```
WITH product_revenue AS (                           -- [②] 실제 제품별 매출 순위 조회
    SELECT p.product_id,
           p.product_name,
           COALESCE((
               SELECT SUM(oi.line_total)
               FROM ecom.order_items oi
               JOIN ecom.orders o ON o.order_id = oi.order_id
               WHERE oi.product_id = p.product_id
                 AND o.order_status IN ('paid', 'shipped', 'delivered')
           ), 0) AS revenue
    FROM ecom.products p
), ranked AS (
    SELECT product_id, product_name, revenue,
           RANK() OVER (ORDER BY revenue DESC) AS revenue_rank
    FROM product_revenue
    WHERE revenue > 0
)
SELECT product_id, product_name, ROUND(revenue, 2) AS revenue, revenue_rank
FROM ranked
WHERE revenue_rank <= 20
ORDER BY revenue_rank, product_id;
```

products 1

WITH product_revenue AS ( SELECT p.product_id, p.prod

| | product_id | product_name | revenue | revenue_rank |
|---|---|---|---|---|
| 1 | 293 | Product 465 | 859,785.6 | 1 |
| 2 | 455 | Product 378 | 430,633.92 | 2 |
| 3 | 501 | Product 439 | 9,189.04 | 3 |
| 4 | 272 | Product 315 | 8,234.57 | 4 |
| 5 | 264 | Product 225 | 8,109.2 | 5 |
| 6 | 528 | Product 549 | 8,040.23 | 6 |
| 7 | 315 | Product 136 | 7,868.31 | 7 |
| 8 | 69 | Product 42 | 7,817.88 | 8 |
| 9 | 235 | Product 114 | 7,584.68 | 9 |
| 10 | 533 | Product 39 | 7,426.7 | 10 |
| 11 | 236 | Product 584 | 7,398.53 | 11 |
| 12 | 63 | Product 32 | 7,375.03 | 12 |
| 13 | 55 | Product 481 | 6,944.38 | 13 |
| 14 | 137 | Product 463 | 6,919.99 | 14 |
| 15 | 33 | Product 211 | 6,859.89 | 15 |
| 16 | 449 | Product 418 | 6,748.8 | 16 |
| 17 | 311 | Product 96 | 6,699.06 | 17 |
| 18 | 181 | Product 264 | 6,586.17 | 18 |
| 19 | 429 | Product 498 | 6,539.14 | 19 |
| 20 | 81 | Product 522 | 6,463.1 | 20 |

Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) WITH product | Enter a SQL expression to filter results (use Ctrl+Space)

| | QUERY PLAN |
|---|---|
| 13 | Buffers: shared hit=70677 |
| 14 | -> Seq Scan on products p (cost=0.00..126896.10 rows=200 width=51) (actual time=1.089..35.299 rows=585 loops=1) |
| 15 | Filter: (COALESCE((SubPlan 2), '0'::numeric) > '0'::numeric) |
| 16 | Rows Removed by Filter: 15 |
| 17 | Buffers: shared hit=70677 |
| 18 | SubPlan 1 |
| 19 | -> Aggregate (cost=158.59..158.60 rows=1 width=32) (actual time=0.025..0.025 rows=1 loops=585) |
| 20 | Buffers: shared hit=35254 |
| 21 | -> Nested Loop (cost=4.70..158.55 rows=14 width=6) (actual time=0.006..0.024 rows=14 loops=585) |
| 22 | Buffers: shared hit=35254 |
| 23 | -> Bitmap Heap Scan on order_items oi (cost=4.42..49.43 rows=17 width=14) (actual time=0.002..0.005 row |
| 24 | Recheck Cond: (product_id = p.product_id) |
| 25 | Heap Blocks: exact=3099 |
| 26 | Buffers: shared hit=4270 |
| 27 | -> Bitmap Index Scan on idx_order_items_product (cost=0.00..4.41 rows=17 width=0) (actual time=0.00 |
| 28 | Index Cond: (product_id = p.product_id) |
| 29 | Buffers: shared hit=1171 |
| 30 | -> Index Scan using orders_pkey on orders o (cost=0.28..6.42 rows=1 width=8) (actual time=0.001..0.001 ro |
| 31 | Index Cond: (order_id = oi.order_id) |
| 32 | Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[])) |
| 33 | Rows Removed by Filter: 0 |
| 34 | Buffers: shared hit=30984 |
| 35 | SubPlan 2 |
| 36 | -> Aggregate (cost=158.59..158.60 rows=1 width=32) (actual time=0.033..0.033 rows=1 loops=600) |
| 37 | Buffers: shared hit=35416 |
| 38 | -> Nested Loop (cost=4.70..158.55 rows=14 width=6) (actual time=0.010..0.031 rows=14 loops=600) |
| 39 | Buffers: shared hit=35416 |
| 40 | -> Bitmap Heap Scan on order_items oi_1 (cost=4.42..49.43 rows=17 width=14) (actual time=0.005..0.009 r |
| 41 | Recheck Cond: (product_id = p.product_id) |
| 42 | Heap Blocks: exact=3114 |
| 43 | Buffers: shared hit=4315 |
| 44 | -> Bitmap Index Scan on idx_order_items_product (cost=0.00..4.41 rows=17 width=0) (actual time=0.00 |
| 45 | Index Cond: (product_id = p.product_id) |
| 46 | Buffers: shared hit=1201 |
| 47 | -> Index Scan using orders_pkey on orders o_1 (cost=0.28..6.42 rows=1 width=8) (actual time=0.001..0.001 |
| 48 | Index Cond: (order_id = oi_1.order_id) |
| 49 | Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[])) |
| 50 | Rows Removed by Filter: 0 |
| 51 | Buffers: shared hit=31101 |
| 52 | Planning: |
| 53 | Buffers: shared hit=24 |
| 54 | Planning Time: 1.451 ms |
| 55 | Execution Time: 37.373 ms |

## 느린 이유와 개선 방법

튜닝 전에는 제품마다 주문상품과 주문을 반복 조회한다. 튜닝 후에는 전체 매출을 제품별로 한 번 집계한 후 Window Function을 적용한다.

Q4 튜닝 후

```
-- Q4. 전체 매출을 한 번만 집계한 뒤 RANK 적용
-- 튜닝: 제품마다 매출을 다시 찾지 않고 전체 데이터를 한 번 읽어 제품별로 묶는다.
-- [① 실행계획 확인용] 반복 SubPlan이 제거됐는지 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)
WITH product_revenue AS (                          -- 전체 데이터를 제품별로 한 번 집계
    SELECT p.product_id,
           p.product_name,
           SUM(oi.line_total) AS revenue
    FROM ecom.orders o
    JOIN ecom.order_items oi ON oi.order_id = o.order_id
    JOIN ecom.products p ON p.product_id = oi.product_id
    WHERE o.order_status IN ('paid', 'shipped', 'delivered')
    GROUP BY p.product_id, p.product_name
), ranked AS (                                     -- 집계 결과에 순위를 부여
    SELECT product_id, product_name, revenue,
           RANK() OVER (ORDER BY revenue DESC) AS revenue_rank -- 매출 내림차순 공동 순위
    FROM product_revenue
)
SELECT product_id, product_name, ROUND(revenue, 2) AS revenue, revenue_rank
FROM ranked
WHERE revenue_rank <= 20
ORDER BY revenue_rank, product_id;
```

```
WITH product_revenue AS (                          -- [②] 실제 제품별 순위 조회
    SELECT p.product_id, p.product_name, SUM(oi.line_total) AS revenue
    FROM ecom.orders o
    JOIN ecom.order_items oi ON oi.order_id = o.order_id
    JOIN ecom.products p ON p.product_id = oi.product_id
    WHERE o.order_status IN ('paid', 'shipped', 'delivered')
    GROUP BY p.product_id, p.product_name
), ranked AS (
    SELECT product_id, product_name, revenue,
           RANK() OVER (ORDER BY revenue DESC) AS revenue_rank
    FROM product_revenue
)
SELECT product_id, product_name, ROUND(revenue, 2) AS revenue, revenue_rank
FROM ranked
WHERE revenue_rank <= 20
ORDER BY revenue_rank, product_id;
```

Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) WITH product | Enter a SQL expression to filter results (use Ctrl+Space)

| | QUERY PLAN |
|---|---|
| 1 | Sort (cost=520.70..522.20 rows=600 width=59) (actual time=9.509..9.513 rows=20 loops=1) |
| 2 | Sort Key: ranked.revenue_rank, ranked.product_id |
| 3 | Sort Method: quicksort Memory: 26kB |
| 4 | Buffers: shared hit=147 |
| 5 | -> Subquery Scan on ranked (cost=481.03..493.01 rows=600 width=59) (actual time=9.481..9.501 rows=20 loops=1) |
| 6 | Buffers: shared hit=147 |
| 7 | -> WindowAgg (cost=481.03..491.51 rows=600 width=59) (actual time=9.479..9.495 rows=20 loops=1) |
| 8 | Run Condition: (rank() OVER (?) <= 20) |
| 9 | Buffers: shared hit=147 |
| 10 | -> Sort (cost=481.01..482.51 rows=600 width=51) (actual time=9.473..9.477 rows=21 loops=1) |
| 11 | Sort Key: product_revenue.revenue DESC |
| 12 | Sort Method: quicksort Memory: 52kB |
| 13 | Buffers: shared hit=147 |
| 14 | -> Subquery Scan on product_revenue (cost=445.83..453.33 rows=600 width=51) (actual time=9.089..9.292 rows=585 loops=1) |
| 15 | Buffers: shared hit=147 |
| 16 | -> HashAggregate (cost=445.83..453.33 rows=600 width=51) (actual time=9.088..9.241 rows=585 loops=1) |
| 17 | Group Key: p.product_id |
| 18 | Batches: 1 Memory Usage: 297kB |
| 19 | Buffers: shared hit=147 |
| 20 | -> Hash Join (cost=151.93..404.63 rows=8240 width=25) (actual time=2.183..7.128 rows=8230 loops=1) |
| 21 | Hash Cond: (oi.product_id = p.product_id) |
| 22 | Buffers: shared hit=147 |
| 23 | -> Hash Join (cost=131.43..362.34 rows=8240 width=14) (actual time=1.843..5.248 rows=8230 loops=1) |
| 24 | Hash Cond: (oi.order_id = o.order_id) |
| 25 | Buffers: shared hit=140 |
| 26 | -> Seq Scan on order_items oi (cost=0.00..203.67 rows=10367 width=22) (actual time=0.010..1.255 rows=10367 loops=1) |
| 27 | Buffers: shared hit=100 |
| 28 | -> Hash (cost=93.08..93.08 rows=3068 width=8) (actual time=1.801..1.802 rows=3068 loops=1) |
| 29 | Buckets: 4096 Batches: 1 Memory Usage: 152kB |
| 30 | Buffers: shared hit=40 |
| 31 | -> Seq Scan on orders o (cost=0.00..93.08 rows=3068 width=8) (actual time=0.012..1.281 rows=3068 loops=1) |
| 32 | Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[])) |
| 33 | Rows Removed by Filter: 792 |
| 34 | Buffers: shared hit=40 |
| 35 | -> Hash (cost=13.00..13.00 rows=600 width=19) (actual time=0.315..0.316 rows=600 loops=1) |
| 36 | Buckets: 1024 Batches: 1 Memory Usage: 39kB |
| 37 | Buffers: shared hit=7 |
| 38 | -> Seq Scan on products p (cost=0.00..13.00 rows=600 width=19) (actual time=0.014..0.147 rows=600 loops=1) |
| 39 | Buffers: shared hit=7 |
| 40 | Planning: |
| 41 | Buffers: shared hit=114 read=3 dirtied=1 |
| 42 | Planning Time: 6.071 ms |
| 43 | Execution Time: 9.654 ms |

products 1

WITH product_revenue AS ( SELECT p.product_id, p.prod | Enter a SQL expression to filter results (use C

| | product_id | product_name | revenue | revenue_rank |
|---|---|---|---|---|
| 1 | 293 | Product 465 | 859,785.6 | 1 |
| 2 | 455 | Product 378 | 430,633.92 | 2 |
| 3 | 501 | Product 439 | 9,189.04 | 3 |
| 4 | 272 | Product 315 | 8,234.57 | 4 |
| 5 | 264 | Product 225 | 8,109.2 | 5 |
| 6 | 528 | Product 549 | 8,040.23 | 6 |
| 7 | 315 | Product 136 | 7,868.31 | 7 |
| 8 | 69 | Product 42 | 7,817.88 | 8 |
| 9 | 235 | Product 114 | 7,584.68 | 9 |
| 10 | 533 | Product 39 | 7,426.7 | 10 |
| 11 | 236 | Product 584 | 7,398.53 | 11 |
| 12 | 63 | Product 32 | 7,375.03 | 12 |
| 13 | 55 | Product 481 | 6,944.38 | 13 |
| 14 | 137 | Product 463 | 6,919.99 | 14 |
| 15 | 33 | Product 211 | 6,859.89 | 15 |
| 16 | 449 | Product 418 | 6,748.8 | 16 |
| 17 | 311 | Product 96 | 6,699.06 | 17 |
| 18 | 181 | Product 264 | 6,586.17 | 18 |
| 19 | 429 | Product 498 | 6,539.14 | 19 |
| 20 | 81 | Product 522 | 6,463.1 | 20 |

| 구분 | 반복 SubPlan | Buffers | Execution Time | 1위 제품/매출 |
|---|---|---|---|---|
| 튜닝 전 | 있음(585/600회) | hit=70,677 | 37.373 ms | Product 465 / 859,785.60 |
| 튜닝 후 | 없음 | hit=147 | 9.654 ms | Product 465 / 859,785.60 |

**결과 해석:** 제품별 반복 SubPlan을 1회 집계로 바꾸어 실행시간이 74.2% 단축되었다.

# 7. Q5 - 고객별 RFM 분석

## 요청사항

고객별 최근 구매 후 경과일(최근성), 실구매 주문 수(빈도), 총매출(금액)을 제공한다.

Q5 튜닝 전

```sql
-- ============================================================
-- Q5. 고객별 RFM
-- 문제점: 고객마다 최근 구매·빈도·금액을 각각 다시 조회한다.
-- 화면에는 M(금액) 상위 20명만 표시한다.
-- 의미: Recency=최근 구매 후 경과일, Frequency=구매 빈도, Monetary=구매 금액이다.
-- ============================================================
-- [① 실행계획 확인용] 고객별 반복 서브쿼리와 loops 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)                  -- 고객별 반복 서브쿼리 비용 측정
SELECT c.customer_id,
       c.full_name,
       CURRENT_DATE - (                              -- 오늘에서 최근 구매일을 뺀 경과일 계산
           SELECT MAX(o.order_ts)::date              -- 고객의 가장 최근 실구매일
           FROM ecom.orders o
           WHERE o.customer_id = c.customer_id
             AND o.order_status IN ('paid', 'shipped', 'delivered')
       ) AS recency_days,                            -- R: 최근 구매 후 경과일
       (
           SELECT COUNT(o.order_id)                  -- 고객의 실구매 주문번호 수
           FROM ecom.orders o
           WHERE o.customer_id = c.customer_id
             AND o.order_status IN ('paid', 'shipped', 'delivered')
       ) AS frequency,                               -- F: 구매 빈도
       (
           SELECT COALESCE(SUM(oi.line_total), 0)    -- 고객의 주문상품 총매출
           FROM ecom.orders o
           JOIN ecom.order_items oi ON oi.order_id = o.order_id
           WHERE o.customer_id = c.customer_id
             AND o.order_status IN ('paid', 'shipped', 'delivered')
       ) AS monetary                                 -- M: 구매 금액
FROM ecom.customers c
WHERE EXISTS (                                       -- 실구매 이력이 있는 고객만 포함
    SELECT 1
    FROM ecom.orders o
    WHERE o.customer_id = c.customer_id
      AND o.order_status IN ('paid', 'shipped', 'delivered')
)
ORDER BY monetary DESC, c.customer_id
LIMIT 20;
```

customers 1

SELECT c.customer_id, c.full_name, CURRENT_DATE - ( SELECT MAX(o.o

| | customer_id | full_name | recency_days | frequency | monetary |
|---|---|---|---|---|---|
| 1 | 8 | Customer 8 | 2 | 23 | 35,154.22 |
| 2 | 16 | Customer 16 | 1 | 23 | 31,003.24 |
| 3 | 18 | Customer 18 | 3 | 23 | 30,283.04 |
| 4 | 14 | Customer 14 | 6 | 23 | 29,322.59 |
| 5 | 6 | Customer 6 | 0 | 22 | 28,703.23 |
| 6 | 13 | Customer 13 | 2 | 22 | 28,633.41 |
| 7 | 2 | Customer 2 | 8 | 23 | 28,623.31 |
| 8 | 1 | Customer 1 | 3 | 23 | 28,546.26 |
| 9 | 20 | Customer 20 | 0 | 22 | 28,531.13 |
| 10 | 9 | Customer 9 | 1 | 23 | 28,488.91 |
| 11 | 17 | Customer 17 | 8 | 23 | 27,317.53 |
| 12 | 24 | Customer 24 | 0 | 22 | 27,075.74 |
| 13 | 23 | Customer 23 | 4 | 22 | 26,717.57 |
| 14 | 30 | Customer 30 | 0 | 23 | 24,815.39 |
| 15 | 4 | Customer 4 | 1 | 23 | 24,726.24 |
| 16 | 12 | Customer 12 | 4 | 23 | 22,419.91 |
| 17 | 19 | Customer 19 | 0 | 23 | 22,412.27 |
| 18 | 22 | Customer 22 | 1 | 23 | 22,206.5 |
| 19 | 7 | Customer 7 | 1 | 23 | 22,034.39 |
| 20 | 29 | Customer 29 | 0 | 22 | 21,696.95 |

```
QUERY PLAN
Limit  (cost=50416.78..51083.45 rows=20 width=65) (actual time=21.113..21.351 rows=20 loops=1)
  Buffers: shared hit=17287
  ->  Result  (cost=50416.78..150418.03 rows=3000 width=65) (actual time=21.113..21.348 rows=20 loops=1)
        Buffers: shared hit=17287
        ->  Sort  (cost=50416.78..50424.28 rows=3000 width=53) (actual time=20.749..20.753 rows=20 loops=1)
              Sort Key: ((SubPlan 4)) DESC, c.customer_id
              Sort Method: top-N heapsort  Memory: 26kB
              Buffers: shared hit=16776
              ->  Hash Join  (cost=159.45..50336.95 rows=3000 width=53) (actual time=2.743..20.024 rows=2217 loops=1)
                    Hash Cond: (c.customer_id = o.customer_id)
                    Buffers: shared hit=16776
                    ->  Seq Scan on customers c  (cost=0.00..70.00 rows=3000 width=21) (actual time=0.033..4.299 rows=3000 loops=1)
                          Buffers: shared hit=40
                    ->  Hash  (cost=126.84..126.84 rows=2609 width=8) (actual time=2.045..2.046 rows=2217 loops=1)
                          Buckets: 4096  Batches: 1  Memory Usage: 119kB
                          Buffers: shared hit=40
                          ->  HashAggregate  (cost=100.75..126.84 rows=2609 width=8) (actual time=1.343..1.516 rows=2217 loops=1)
                                Group Key: o.customer_id
                                Batches: 1  Memory Usage: 241kB
                                Buffers: shared hit=40
                                ->  Seq Scan on orders o  (cost=0.00..93.08 rows=3068 width=8) (actual time=0.012..0.811 rows=3068 loops=1)
                                      Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                                      Rows Removed by Filter: 792
                                      Buffers: shared hit=40
              SubPlan 4
                ->  Aggregate  (cost=16.68..16.69 rows=1 width=32) (actual time=0.006..0.006 rows=1 loops=2217)
                      Buffers: shared hit=16696
                      ->  Nested Loop  (cost=0.57..16.67 rows=3 width=6) (actual time=0.003..0.005 rows=4 loops=2217)
                            Buffers: shared hit=16696
                            ->  Index Scan using idx_orders_customer_ts on orders o_3  (cost=0.28..8.30 rows=1 width=8) (actual time=0.002..0.002 rows=1 loops=2217)
                                  Index Cond: (customer_id = c.customer_id)
                                  Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                                  Rows Removed by Filter: 0
                                  Buffers: shared hit=7438
                            ->  Index Scan using idx_order_items_order on order_items oi  (cost=0.29..8.34 rows=3 width=14) (actual time=0.001..0.002 rows=3 loops=3068)
                                  Index Cond: (order_id = o_3.order_id)
                                  Buffers: shared hit=9258
              SubPlan 2
                ->  Result  (cost=8.30..8.31 rows=1 width=4) (actual time=0.003..0.003 rows=1 loops=20)
                      Buffers: shared hit=60
                      InitPlan 1
                        ->  Limit  (cost=0.28..8.30 rows=1 width=8) (actual time=0.003..0.003 rows=1 loops=20)
                              Buffers: shared hit=60
                              ->  Index Scan using idx_orders_customer_ts on orders o_1  (cost=0.28..8.30 rows=1 width=8) (actual time=0.002..0.002 rows=1 loops=20)
                                    Index Cond: (customer_id = c.customer_id)
                                    Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                                    Buffers: shared hit=60
              SubPlan 3
                ->  Aggregate  (cost=8.30..8.31 rows=1 width=8) (actual time=0.025..0.025 rows=1 loops=20)
                      Buffers: shared hit=451
                      ->  Index Scan using idx_orders_customer_ts on orders o_2  (cost=0.28..8.30 rows=1 width=8) (actual time=0.016..0.023 rows=23 loops=20)
                            Index Cond: (customer_id = c.customer_id)
                            Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                            Rows Removed by Filter: 0
                            Buffers: shared hit=451
Planning:
  Buffers: shared hit=27
Planning Time: 3.302 ms
Execution Time: 21.915 ms
```

## 느린 이유와 개선 방법

튜닝 전에는 고객마다 최근 구매·빈도·금액을 각각 조회한다. 튜닝 후에는 주문별 금액을 선집계하고 고객별 `MAX`, `COUNT`, `SUM`을 한 번에 계산한다.

Q5 튜닝 후

```sql
-- Q5. 주문별 매출 선집계 후 고객별 한 번의 GROUP BY로 RFM 산출
-- 튜닝: 고객 한 명마다 세 가지 서브쿼리를 반복하는 작업을 제거한다.
-- [① 실행계획 확인용] 고객별 반복 조회 대신 한 번에 집계하는지 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)
WITH order_totals AS (                          -- 주문별 금액을 먼저 집계
    SELECT o.order_id, o.customer_id, o.order_ts,
           SUM(oi.line_total) AS order_total
    FROM ecom.orders o
    JOIN ecom.order_items oi ON oi.order_id = o.order_id
    WHERE o.order_status IN ('paid', 'shipped', 'delivered')
    GROUP BY o.order_id, o.customer_id, o.order_ts
), rfm AS (                                     -- 주문별 결과를 고객별 RFM으로 재집계
    SELECT customer_id,
           CURRENT_DATE - MAX(order_ts)::date AS recency_days, -- 최근 구매 경과일
           COUNT(order_id) AS frequency,        -- 구매 주문번호 수
           SUM(order_total) AS monetary         -- 고객 총매출
    FROM order_totals
    GROUP BY customer_id
)
SELECT c.customer_id, c.full_name,
       r.recency_days, r.frequency, ROUND(r.monetary, 2) AS monetary
FROM rfm r
JOIN ecom.customers c ON c.customer_id = r.customer_id
ORDER BY r.monetary DESC, c.customer_id
LIMIT 20;
```

Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) WITH order_totals AS ( SELEC

| | QUERY PLAN |
|---|---|
| 1 | Limit (cost=595.46..595.51 rows=20 width=97) (actual time=13.181..13.185 rows=20 loops=1) |
| 2 | Buffers: shared hit=180 |
| 3 | -> Sort (cost=595.46..595.96 rows=200 width=97) (actual time=13.180..13.183 rows=20 loops=1) |
| 4 | Sort Key: r.monetary DESC, c.customer_id |
| 5 | Sort Method: top-N heapsort Memory: 28kB |
| 6 | Buffers: shared hit=180 |
| 7 | -> Hash Join (cost=511.75..590.14 rows=200 width=97) (actual time=11.699..12.710 rows=2217 loops=1) |
| 8 | Hash Cond: (c.customer_id = r.customer_id) |
| 9 | Buffers: shared hit=180 |
| 10 | -> Seq Scan on customers c (cost=0.00..70.00 rows=3000 width=21) (actual time=0.010..0.324 rows=3000 loops=1) |
| 11 | Buffers: shared hit=40 |
| 12 | -> Hash (cost=509.25..509.25 rows=200 width=52) (actual time=11.675..11.677 rows=2217 loops=1) |
| 13 | Buckets: 4096 (originally 1024) Batches: 1 (originally 1) Memory Usage: 171kB |
| 14 | Buffers: shared hit=140 |
| 15 | -> Subquery Scan on r (cost=503.25..509.25 rows=200 width=52) (actual time=10.032..11.245 rows=2217 loops=1) |
| 16 | Buffers: shared hit=140 |
| 17 | -> HashAggregate (cost=503.25..507.25 rows=200 width=52) (actual time=10.031..11.010 rows=2217 loops=1) |
| 18 | Group Key: o.customer_id |
| 19 | Batches: 1 Memory Usage: 1153kB |
| 20 | Buffers: shared hit=140 |
| 21 | -> HashAggregate (cost=403.54..441.89 rows=3068 width=56) (actual time=7.888..8.841 rows=3068 loops= |
| 22 | Group Key: o.order_id |
| 23 | Batches: 1 Memory Usage: 1393kB |
| 24 | Buffers: shared hit=140 |
| 25 | -> Hash Join (cost=131.43..362.34 rows=8240 width=30) (actual time=1.422..5.116 rows=8230 loops=1) |
| 26 | Hash Cond: (oi.order_id = o.order_id) |
| 27 | Buffers: shared hit=140 |
| 28 | -> Seq Scan on order_items oi (cost=0.00..203.67 rows=10367 width=14) (actual time=0.013..1.160 ro |
| 29 | Buffers: shared hit=100 |
| 30 | -> Hash (cost=93.08..93.08 rows=3068 width=24) (actual time=1.389..1.389 rows=3068 loops=1) |
| 31 | Buckets: 4096 Batches: 1 Memory Usage: 200kB |
| 32 | Buffers: shared hit=40 |
| 33 | -> Seq Scan on orders o (cost=0.00..93.08 rows=3068 width=24) (actual time=0.016..0.940 rows= |
| 34 | Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[])) |
| 35 | Rows Removed by Filter: 792 |
| 36 | Buffers: shared hit=40 |
| 37 | Planning: |
| 38 | Buffers: shared hit=18 |
| 39 | Planning Time: 0.729 ms |
| 40 | Execution Time: 13.785 ms |

```sql
WITH order_totals AS (                          -- [②] 실제 고객별 RFM 조회
    SELECT o.order_id, o.customer_id, o.order_ts,
           SUM(oi.line_total) AS order_total
    FROM ecom.orders o
    JOIN ecom.order_items oi ON oi.order_id = o.order_id
    WHERE o.order_status IN ('paid', 'shipped', 'delivered')
    GROUP BY o.order_id, o.customer_id, o.order_ts
), rfm AS (
    SELECT customer_id, CURRENT_DATE - MAX(order_ts)::date AS recency_days,
           COUNT(order_id) AS frequency, SUM(order_total) AS monetary
    FROM order_totals
    GROUP BY customer_id
)
SELECT c.customer_id, c.full_name,
       r.recency_days, r.frequency, ROUND(r.monetary, 2) AS monetary
FROM rfm r
JOIN ecom.customers c ON c.customer_id = r.customer_id
ORDER BY r.monetary DESC, c.customer_id
LIMIT 20;
```

customers 1

WITH order_totals AS ( SELECT o.order_id, o.customer_id, o.order_ts, SUI

| | customer_id | full_name | recency_days | frequency | monetary |
|---|---|---|---|---|---|
| 1 | 8 | Customer 8 | 2 | 23 | 35,154.22 |
| 2 | 16 | Customer 16 | 1 | 23 | 31,003.24 |
| 3 | 18 | Customer 18 | 3 | 23 | 30,283.04 |
| 4 | 14 | Customer 14 | 6 | 23 | 29,322.59 |
| 5 | 6 | Customer 6 | 0 | 22 | 28,703.23 |
| 6 | 13 | Customer 13 | 2 | 22 | 28,633.41 |
| 7 | 2 | Customer 2 | 8 | 23 | 28,623.31 |
| 8 | 1 | Customer 1 | 3 | 23 | 28,546.26 |
| 9 | 20 | Customer 20 | 0 | 22 | 28,531.13 |
| 10 | 9 | Customer 9 | 1 | 23 | 28,488.91 |
| 11 | 17 | Customer 17 | 8 | 23 | 27,317.53 |
| 12 | 24 | Customer 24 | 0 | 22 | 27,075.74 |
| 13 | 23 | Customer 23 | 4 | 22 | 25,717.57 |
| 14 | 30 | Customer 30 | 0 | 23 | 24,815.39 |
| 15 | 4 | Customer 4 | 1 | 23 | 24,726.24 |
| 16 | 12 | Customer 12 | 4 | 23 | 22,419.91 |
| 17 | 19 | Customer 19 | 0 | 23 | 22,412.27 |
| 18 | 22 | Customer 22 | 1 | 23 | 22,206.5 |
| 19 | 7 | Customer 7 | 1 | 23 | 22,034.39 |
| 20 | 29 | Customer 29 | 0 | 22 | 21,696.95 |

| 구분 | 집계 방식 | Buffers | Execution Time | 전후 결과 동일 여부 |
|---|---|---|---|---|
| 튜닝 전 | 고객별 상관 서브쿼리 | hit=17,287 | 21.915 ms | - |
| 튜닝 후 | 주문 선집계 후 고객 GROUP BY | hit=180 | 13.785 ms | 동일 |

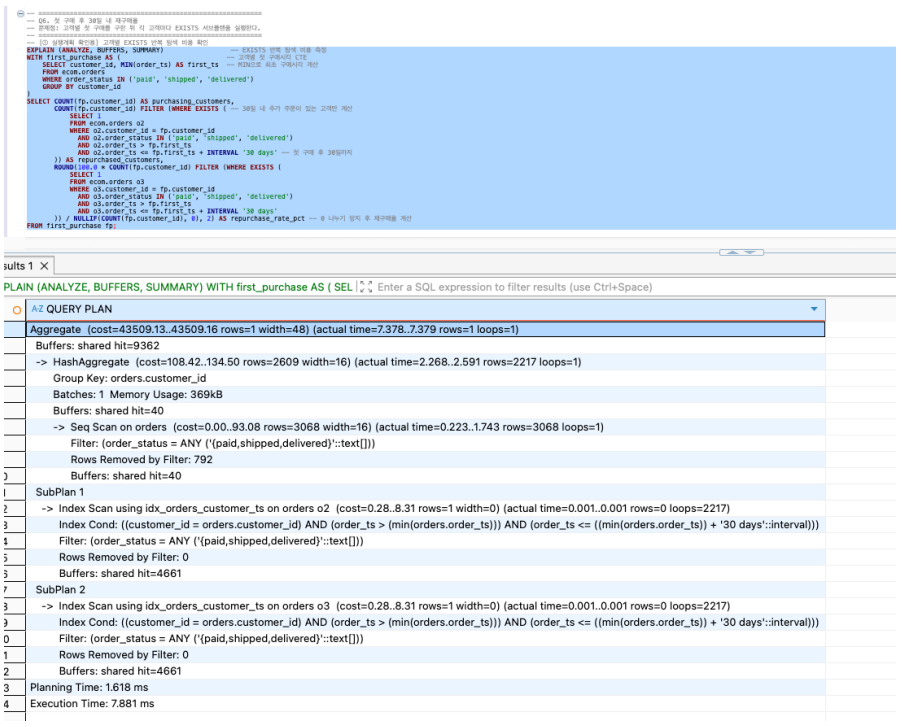
**RFM 결과 해석:** Customer 8이 최근성 2일·빈도 23회·금액 35,154.22로 최고 금액 고객이며 37.1% 단축되었다.

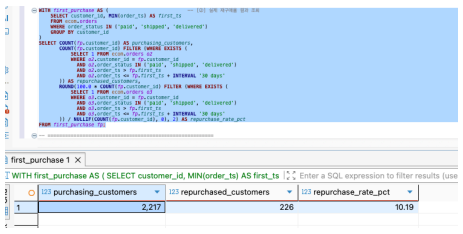
# 8. Q6 - 첫 구매 후 30일 내 재구매율

## 요청사항

실구매 고객 중 첫 구매 후 30일 안에 추가 실구매가 있었던 고객의 비율을 제공한다.

Q6 튜닝 전

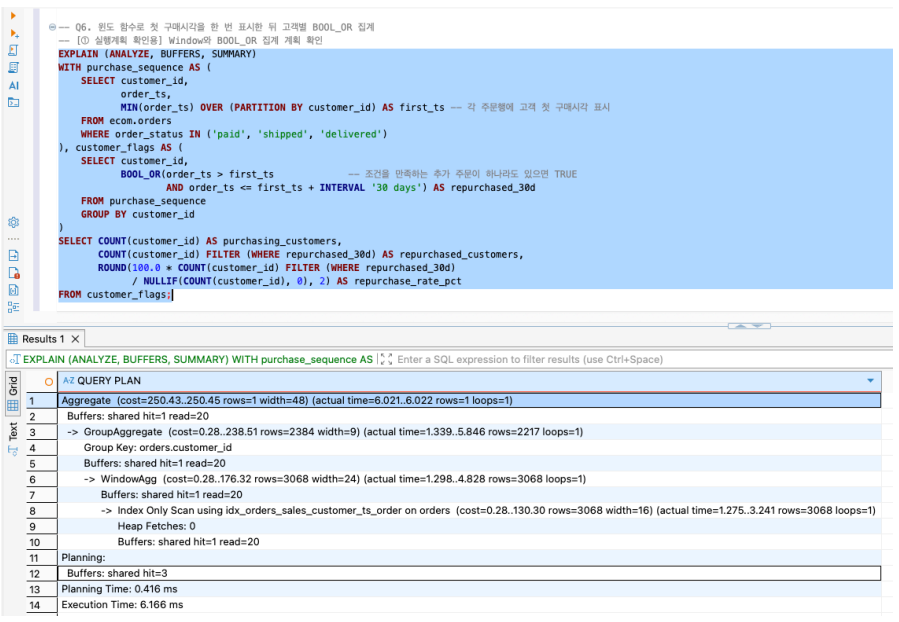


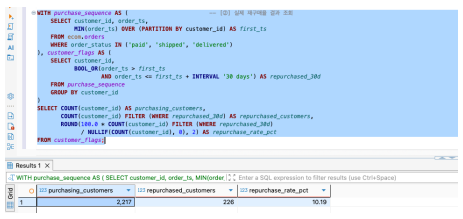


## 느린 이유와 개선 방법

튜닝 전에는 고객별 첫 구매 이후 주문을 `EXISTS`로 반복 탐색한다. 튜닝 후에는 Window Function으로 첫 구매시각을 표시하고 고객별 `BOOL_OR`로 재구매 여부를 한 번 집계한다.

Q6 튜닝 후





| 구분 | 구매고객 | 30일 내 재구매고객 | 재구매율 | Execution Time |
|---|---|---|---|---|
| 튜닝 전 | 2,217 | 226 | 10.19% | 7.881 ms |
| 튜닝 후 | 2,217 | 226 | 10.19% | 6.166 ms |

**결과 해석:** `재구매율`은 전후 동일하며 Window와 BOOL_OR 집계로 실행시간이 21.8% 단축되었다.

# 9. Q7 - 재주문 임계치 미만 상품

## 요청사항

`qty_on_hand < reorder_point`인 상품과 부족수량을 제공한다.

Q7 튜닝 전

```
-- ==========================================================
-- Q7. 재고가 재주문 임계치보다 낮은 상품
-- 문제점: 각 행마다 같은 inventory 행의 임계치를 상관 조회한다.
-- ==========================================================
-- [① 실행계획 확인용] inventory 반복 조회와 Scan 방식 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)                  -- 재고 테이블 접근 방식과 비용 측정
SELECT p.product_id, p.sku, p.product_name,
       i.qty_on_hand, i.reorder_point,               -- 현재고와 재주문 기준 수량
       i.reorder_point - i.qty_on_hand AS shortage_qty -- 보충이 필요한 부족수량
FROM ecom.products p
JOIN ecom.inventory i ON i.product_id = p.product_id
WHERE i.qty_on_hand < (                              -- 현재고가 재주문 기준보다 낮은 행
    SELECT MAX(i2.reorder_point)                     -- 같은 재고행의 기준을 다시 조회
    FROM ecom.inventory i2
    WHERE i2.product_id = i.product_id
)
ORDER BY shortage_qty DESC, p.product_id
LIMIT 20;
```

Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) SELECT p.proi | Enter a SQL expression to filter results (use Ctrl+Space)

```
QUERY PLAN
Limit  (cost=5022.35..5022.40 rows=20 width=42) (actual time=1.923..1.927 rows=20 loops=1)
  Buffers: shared hit=1912 dirtied=4
  ->  Sort  (cost=5022.35..5022.85 rows=200 width=42) (actual time=1.922..1.925 rows=20 loops=1)
        Sort Key: ((i.reorder_point - i.qty_on_hand)) DESC, p.product_id
        Sort Method: top-N heapsort  Memory: 27kB
        Buffers: shared hit=1912 dirtied=4
        ->  Hash Join  (cost=20.50..5017.02 rows=200 width=42) (actual time=0.684..1.874 rows=66 loops=1)
              Hash Cond: (i.product_id = p.product_id)
              Buffers: shared hit=1912 dirtied=4
              ->  Seq Scan on inventory i  (cost=0.00..4995.50 rows=200 width=16) (actual time=0.058..1.227 rows=66 loops=1)
                    Filter: (qty_on_hand < (SubPlan 1))
                    Rows Removed by Filter: 534
                    Buffers: shared hit=1905 dirtied=4
                    SubPlan 1
                      ->  Aggregate  (cost=8.29..8.30 rows=1 width=4) (actual time=0.002..0.002 rows=1 loops=600)
                            Buffers: shared hit=1900 dirtied=2
                            ->  Index Scan using inventory_pkey on inventory i2  (cost=0.28..8.29 rows=1 width=4) (actual time=0.001..0.001 rows=1 loops=600)
                                  Index Cond: (product_id = i.product_id)
                                  Buffers: shared hit=1900 dirtied=2
              ->  Hash  (cost=13.00..13.00 rows=600 width=30) (actual time=0.598..0.599 rows=600 loops=1)
                    Buckets: 1024  Batches: 1  Memory Usage: 45kB
                    Buffers: shared hit=7
                    ->  Seq Scan on products p  (cost=0.00..13.00 rows=600 width=30) (actual time=0.005..0.456 rows=600 loops=1)
                          Buffers: shared hit=7
Planning:
  Buffers: shared hit=42 dirtied=4
Planning Time: 4.890 ms
Execution Time: 2.017 ms
```

```
SELECT p.product_id, p.sku, p.product_name,          -- [②] 실제 부족 재고 상품 조회
       i.qty_on_hand, i.reorder_point,
       i.reorder_point - i.qty_on_hand AS shortage_qty
FROM ecom.products p
JOIN ecom.inventory i ON i.product_id = p.product_id
WHERE i.qty_on_hand < (
    SELECT MAX(i2.reorder_point) FROM ecom.inventory i2
    WHERE i2.product_id = i.product_id
)
ORDER BY shortage_qty DESC, p.product_id
LIMIT 20;
-- ==========================================================
```

products(+) 1

SELECT p.product_id, p.sku, p.product_name, i.qty_on_hi | Enter a SQL expression to filter results (use Ctrl+Space)

| | product_id | sku | product_name | qty_on_hand | reorder_point | shortage_qty |
|---|---|---|---|---|---|---|
| 1 | 320 | SKU-000186 | Product 186 | 0 | 30 | 30 |
| 2 | 454 | SKU-000428 | Product 428 | 0 | 30 | 30 |
| 3 | 102 | SKU-000022 | Product 22 | 1 | 30 | 29 |
| 4 | 136 | SKU-000163 | Product 163 | 1 | 30 | 29 |
| 5 | 188 | SKU-000594 | Product 594 | 1 | 30 | 29 |
| 6 | 277 | SKU-000015 | Product 15 | 1 | 30 | 29 |
| 7 | 596 | SKU-000590 | Product 590 | 1 | 30 | 29 |
| 8 | 226 | SKU-000374 | Product 374 | 2 | 30 | 28 |
| 9 | 387 | SKU-000447 | Product 447 | 2 | 30 | 28 |
| 10 | 520 | SKU-000519 | Product 519 | 2 | 30 | 28 |
| 11 | 540 | SKU-000359 | Product 359 | 2 | 30 | 28 |
| 12 | 276 | SKU-000135 | Product 135 | 3 | 30 | 27 |
| 13 | 335 | SKU-000336 | Product 336 | 3 | 30 | 27 |
| 14 | 25 | SKU-000231 | Product 231 | 4 | 30 | 26 |
| 15 | 203 | SKU-000404 | Product 404 | 4 | 30 | 26 |
| 16 | 456 | SKU-000578 | Product 578 | 4 | 30 | 26 |
| 17 | 485 | SKU-000279 | Product 279 | 4 | 30 | 26 |
| 18 | 498 | SKU-000429 | Product 429 | 4 | 30 | 26 |
| 19 | 530 | SKU-000559 | Product 559 | 4 | 30 | 26 |
| 20 | 159 | SKU-000373 | Product 373 | 5 | 30 | 25 |

## 느린 이유와 개선 방법

튜닝 전에는 같은 재고 행의 임계치를 다시 상관 조회한다. 튜닝 후에는 직접 비교 조건과 저재고 행만 포함하는 부분 인덱스를 사용한다.

Q7 튜닝 후

```
-- Q7. 직접 조건 + 저재고 부분 인덱스
-- 튜닝: 저재고 행만 저장한 부분 인덱스를 이용해 필요한 후보를 빠르게 찾는다.
-- [① 실행계획 확인용] 저재고 부분 인덱스 사용 여부 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)
SELECT p.product_id, p.sku, p.product_name,
       i.qty_on_hand, i.reorder_point,
       i.reorder_point - i.qty_on_hand AS shortage_qty
FROM ecom.inventory i
JOIN ecom.products p ON p.product_id = i.product_id
WHERE i.qty_on_hand < i.reorder_point                -- 현재고가 재주문 기준보다 낮은 행만 필터
ORDER BY shortage_qty DESC, p.product_id
LIMIT 20;
```

Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) SELECT p.product_id, p.sku, p Enter a SQL expression to filter results (use Ctrl+Space)

```
QUERY PLAN
Limit  (cost=35.41..35.46 rows=20 width=42) (actual time=0.392..0.399 rows=20 loops=1)
  Buffers: shared hit=12
  ->  Sort  (cost=35.41..35.91 rows=200 width=42) (actual time=0.391..0.395 rows=20 loops=1)
        Sort Key: ((i.reorder_point - i.qty_on_hand)) DESC, p.product_id
        Sort Method: top-N heapsort  Memory: 27kB
        Buffers: shared hit=12
        ->  Hash Join  (cost=15.00..30.09 rows=200 width=42) (actual time=0.154..0.351 rows=66 loops=1)
              Hash Cond: (p.product_id = i.product_id)
              Buffers: shared hit=12
              ->  Seq Scan on products p  (cost=0.00..13.00 rows=600 width=30) (actual time=0.010..0.094 rows=600 loops=1)
                    Buffers: shared hit=7
              ->  Hash  (cost=12.50..12.50 rows=200 width=16) (actual time=0.132..0.132 rows=66 loops=1)
                    Buckets: 1024  Batches: 1  Memory Usage: 12kB
                    Buffers: shared hit=5
                    ->  Seq Scan on inventory i  (cost=0.00..12.50 rows=200 width=16) (actual time=0.011..0.112 rows=66 loops=1)
                          Filter: (qty_on_hand < reorder_point)
                          Rows Removed by Filter: 534
                          Buffers: shared hit=5
Planning:
  Buffers: shared hit=12
Planning Time: 0.379 ms
Execution Time: 0.428 ms
```

```
SELECT p.product_id, p.sku, p.product_name,          -- [②] 실제 부족 재고 상품 조회
       i.qty_on_hand, i.reorder_point,
       i.reorder_point - i.qty_on_hand AS shortage_qty
FROM ecom.inventory i
JOIN ecom.products p ON p.product_id = i.product_id
WHERE i.qty_on_hand < i.reorder_point
ORDER BY shortage_qty DESC, p.product_id
LIMIT 20;
```

products(+) 1

SELECT p.product_id, p.sku, p.product_name, i.qty_on_hand, i.reorder_pi Enter a SQL expression to filter results (use Ctrl+Space)

| | product_id | sku | product_name | qty_on_hand | reorder_point | shortage_qty |
|---|---|---|---|---|---|---|
| 1 | 320 | SKU-000186 | Product 186 | 0 | 30 | 30 |
| 2 | 454 | SKU-000428 | Product 428 | 0 | 30 | 30 |
| 3 | 102 | SKU-000022 | Product 22 | 1 | 30 | 29 |
| 4 | 136 | SKU-000163 | Product 163 | 1 | 30 | 29 |
| 5 | 188 | SKU-000594 | Product 594 | 1 | 30 | 29 |
| 6 | 277 | SKU-000015 | Product 15 | 1 | 30 | 29 |
| 7 | 596 | SKU-000590 | Product 590 | 1 | 30 | 29 |
| 8 | 226 | SKU-000374 | Product 374 | 2 | 30 | 28 |
| 9 | 387 | SKU-000447 | Product 447 | 2 | 30 | 28 |
| 10 | 520 | SKU-000519 | Product 519 | 2 | 30 | 28 |
| 11 | 540 | SKU-000359 | Product 359 | 2 | 30 | 28 |
| 12 | 276 | SKU-000135 | Product 135 | 3 | 30 | 27 |
| 13 | 335 | SKU-000336 | Product 336 | 3 | 30 | 27 |
| 14 | 25 | SKU-000231 | Product 231 | 4 | 30 | 26 |
| 15 | 203 | SKU-000404 | Product 404 | 4 | 30 | 26 |
| 16 | 456 | SKU-000578 | Product 578 | 4 | 30 | 26 |
| 17 | 485 | SKU-000279 | Product 279 | 4 | 30 | 26 |
| 18 | 498 | SKU-000429 | Product 429 | 4 | 30 | 26 |
| 19 | 530 | SKU-000559 | Product 559 | 4 | 30 | 26 |
| 20 | 159 | SKU-000373 | Product 373 | 5 | 30 | 25 |

| 구분 | Inventory 접근 방식 | 반복 횟수 | Buffers | Execution Time |
|---|---|---|---|---|
| 튜닝 전 | Seq Scan + SubPlan | 600회 | hit=1,912 | 2.017 ms |
| 튜닝 후 | Seq Scan(직접 비교) | 없음 | hit=12 | 0.428 ms |

**부족 상품 결과 해석:** 저재고 상품은 66개이고 최대 부족수량은 30개이며 실행시간은 78.8% 단축되었다.

# 10. Q8 - 리뷰 기반 효자상품

## 요청사항

평균 평점 4.5 이상이고 리뷰가 50개 이상인 제품을 제공한다.

Q8 튜닝 전

```sql
-- ==============================================================
-- Q8. 평점 4.5 이상, 리뷰 50개 이상 효자상품
-- 문제점: 상품마다 AVG와 COUNT를 SELECT와 WHERE에서 반복 계산한다.
-- ==============================================================
-- [① 실행계획 확인용] 제품별 리뷰 반복 집계 횟수 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)                    -- 반복되는 리뷰 서브쿼리 비용 측정
SELECT p.product_id,
       p.product_name,
       ROUND((SELECT AVG(r.rating) FROM ecom.reviews r
              WHERE r.product_id = p.product_id), 2) AS avg_rating, -- 제품별 평균평점
       (SELECT COUNT(r.review_id) FROM ecom.reviews r
        WHERE r.product_id = p.product_id) AS review_count -- 제품별 리뷰 수
FROM ecom.products p
WHERE (SELECT AVG(r.rating) FROM ecom.reviews r      -- 평균평점을 다시 계산
       WHERE r.product_id = p.product_id) >= 4.5
  AND (SELECT COUNT(r.review_id) FROM ecom.reviews r -- 리뷰 수를 다시 계산
       WHERE r.product_id = p.product_id) >= 50
ORDER BY avg_rating DESC, review_count DESC, p.product_id;
```

```sql
SELECT p.product_id,                                   -- [①] 실제 효자상품 결과 조회
       p.product_name,
       ROUND((SELECT AVG(r.rating) FROM ecom.reviews r
              WHERE r.product_id = p.product_id), 2) AS avg_rating,
       (SELECT COUNT(r.review_id) FROM ecom.reviews r
        WHERE r.product_id = p.product_id) AS review_count
FROM ecom.products p
WHERE (SELECT AVG(r.rating) FROM ecom.reviews r
       WHERE r.product_id = p.product_id) >= 4.5
  AND (SELECT COUNT(r.review_id) FROM ecom.reviews r
       WHERE r.product_id = p.product_id) >= 50
ORDER BY avg_rating DESC, review_count DESC, p.product_id;
-- ==============================================================
```

products 1

SELECT p.product_id, p.product_name, ROUND((SELECT AVG(r.rating) FF

| | product_id | product_name | avg_rating | review_count |
|---|---|---|---|---|
| 1 | 57 | Product 491 | 4.85 | 160 |
| 2 | 229 | Product 464 | 4.83 | 160 |
| 3 | 163 | Product 253 | 4.81 | 160 |
| 4 | 263 | Product 285 | 4.79 | 160 |
| 5 | 426 | Product 258 | 4.79 | 160 |
| 6 | 243 | Product 395 | 4.78 | 160 |
| 7 | 268 | Product 25 | 4.77 | 160 |
| 8 | 297 | Product 75 | 4.76 | 160 |
| 9 | 481 | Product 189 | 4.76 | 160 |
| 10 | 356 | Product 556 | 4.75 | 161 |
| 11 | 178 | Product 113 | 4.74 | 160 |
| 12 | 114 | Product 262 | 4.69 | 160 |

Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) SELECT p.product_id, p.produ

```
QUERY PLAN
        Buffers: shared hit=66
        -> Bitmap Heap Scan on reviews r  (cost=4.40..25.39 rows=16 width=4) (actual time=0.011..0.022 rows=160 loops=12)
           Recheck Cond: (product_id = p.product_id)
           Heap Blocks: exact=32
           Buffers: shared hit=66
           -> Bitmap Index Scan on reviews_product_id_customer_id_key  (cost=0.00..4.40 rows=16 width=0) (actual time=0.009..0.009 rows=160 loops=12)
              Index Cond: (product_id = p.product_id)
              Buffers: shared hit=34
    SubPlan 2
      -> Aggregate  (cost=25.43..25.44 rows=1 width=8) (actual time=0.034..0.034 rows=1 loops=12)
        Buffers: shared hit=66
        -> Bitmap Heap Scan on reviews r_1  (cost=4.40..25.39 rows=16 width=8) (actual time=0.011..0.022 rows=160 loops=12)
           Recheck Cond: (product_id = p.product_id)
           Heap Blocks: exact=32
           Buffers: shared hit=66
           -> Bitmap Index Scan on reviews_product_id_customer_id_key  (cost=0.00..4.40 rows=16 width=0) (actual time=0.008..0.008 rows=160 loops=12)
              Index Cond: (product_id = p.product_id)
              Buffers: shared hit=34
    SubPlan 4
      -> Aggregate  (cost=25.43..25.44 rows=1 width=8) (actual time=0.005..0.005 rows=1 loops=600)
        Buffers: shared hit=1360
        -> Bitmap Heap Scan on reviews r_3  (cost=4.40..25.39 rows=16 width=8) (actual time=0.003..0.003 rows=3 loops=600)
           Recheck Cond: (product_id = p.product_id)
           Heap Blocks: exact=150
           Buffers: shared hit=1360
           -> Bitmap Index Scan on reviews_product_id_customer_id_key  (cost=0.00..4.40 rows=16 width=0) (actual time=0.002..0.002 rows=3 loops=600)
              Index Cond: (product_id = p.product_id)
              Buffers: shared hit=1210
    SubPlan 3
      -> Aggregate  (cost=25.43..25.44 rows=1 width=32) (actual time=0.041..0.041 rows=1 loops=12)
        Buffers: shared hit=66
        -> Bitmap Heap Scan on reviews r_2  (cost=4.40..25.39 rows=16 width=4) (actual time=0.012..0.024 rows=160 loops=12)
           Recheck Cond: (product_id = p.product_id)
           Heap Blocks: exact=32
           Buffers: shared hit=66
           -> Bitmap Index Scan on reviews_product_id_customer_id_key  (cost=0.00..4.40 rows=16 width=0) (actual time=0.009..0.009 rows=160 loops=12)
              Index Cond: (product_id = p.product_id)
              Buffers: shared hit=34
Planning:
 Buffers: shared hit=206 dirtied=7
Planning Time: 4.214 ms
Execution Time: 4.551 ms
```

## 느린 이유와 개선 방법

튜닝 전에는 제품별 평균과 건수를 SELECT와 WHERE에서 반복 계산한다. 튜닝 후에는 리뷰를 한 번 읽고 `GROUP BY`와 `HAVING`으로 조건을 적용한다.

Q8 튜닝 후

```sql
-- Q8. 리뷰 테이블을 한 번만 읽고 GROUP BY/HAVING 적용
-- 튜닝: 제품별 AVG와 COUNT를 한 번 계산하고 HAVING에서 두 조건을 함께 적용한다.
-- [① 실행계획 확인용] reviews를 한 번만 집계하는지 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)
SELECT p.product_id,
       p.product_name,
       ROUND(AVG(r.rating), 2) AS avg_rating,        -- 제품별 평균평점
       COUNT(r.review_id) AS review_count            -- 제품별 리뷰번호 수
FROM ecom.reviews r
JOIN ecom.products p ON p.product_id = r.product_id
GROUP BY p.product_id, p.product_name
HAVING AVG(r.rating) >= 4.5                          -- 집계 후 평균평점 조건 적용
   AND COUNT(r.review_id) >= 50                      -- 집계 후 리뷰 수 조건 적용
ORDER BY avg_rating DESC, review_count DESC, p.product_id;
```

```sql
SELECT p.product_id, p.product_name,                 -- [②] 실제 효자상품 결과 조회
       ROUND(AVG(r.rating), 2) AS avg_rating,
       COUNT(r.review_id) AS review_count
FROM ecom.reviews r
JOIN ecom.products p ON p.product_id = r.product_id
GROUP BY p.product_id, p.product_name
HAVING AVG(r.rating) >= 4.5
   AND COUNT(r.review_id) >= 50
ORDER BY avg_rating DESC, review_count DESC, p.product_id;
```

products 1

SELECT p.product_id, p.product_name, ROUND(AVG(r.rating), 2) AS avg_

| | product_id | product_name | avg_rating | review_count |
|---|---|---|---|---|
| 1 | 57 | Product 491 | 4.85 | 160 |
| 2 | 229 | Product 464 | 4.83 | 160 |
| 3 | 163 | Product 253 | 4.81 | 160 |
| 4 | 263 | Product 285 | 4.79 | 160 |
| 5 | 426 | Product 258 | 4.79 | 160 |
| 6 | 243 | Product 395 | 4.78 | 160 |
| 7 | 268 | Product 25 | 4.77 | 160 |
| 8 | 297 | Product 75 | 4.76 | 160 |
| 9 | 481 | Product 189 | 4.76 | 160 |
| 10 | 356 | Product 556 | 4.75 | 161 |
| 11 | 178 | Product 113 | 4.74 | 160 |
| 12 | 114 | Product 262 | 4.69 | 160 |

Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) SELECT p.product_id, p.produ

```
QUERY PLAN
Sort  (cost=95.27..95.44 rows=67 width=59) (actual time=0.946..0.947 rows=12 loops=1)
  Sort Key: (round(avg(r.rating), 2)) DESC, (count(r.review_id)) DESC, p.product_id
  Sort Method: quicksort  Memory: 25kB
  Buffers: shared hit=28
  -> HashAggregate  (cost=82.57..93.24 rows=67 width=59) (actual time=0.908..0.935 rows=12 loops=1)
     Group Key: p.product_id
     Filter: ((avg(r.rating) >= 4.5) AND (count(r.review_id) >= 50))
     Batches: 1  Memory Usage: 73kB
     Rows Removed by Filter: 118
     Buffers: shared hit=28
     -> Hash Join  (cost=20.50..67.28 rows=2039 width=31) (actual time=0.174..0.667 rows=2039 loops=1)
        Hash Cond: (r.product_id = p.product_id)
        Buffers: shared hit=28
        -> Seq Scan on reviews r  (cost=0.00..41.39 rows=2039 width=20) (actual time=0.012..0.225 rows=2039 loops=1)
           Buffers: shared hit=21
        -> Hash  (cost=13.00..13.00 rows=600 width=19) (actual time=0.149..0.149 rows=600 loops=1)
           Buckets: 1024  Batches: 1  Memory Usage: 39kB
           Buffers: shared hit=7
           -> Seq Scan on products p  (cost=0.00..13.00 rows=600 width=19) (actual time=0.006..0.075 rows=600 loops=1)
              Buffers: shared hit=7
Planning:
 Buffers: shared hit=31 read=4
Planning Time: 2.005 ms
Execution Time: 1.016 ms
```

| 구분 | Reviews 접근 방식 | Buffers | Execution Time | 조건 충족 상품 수 |
|---|---|---|---|---|
| 튜닝 전 | 반복 Bitmap Scan | hit=1,565 | 4.551 ms | 12개 |
| 튜닝 후 | Seq Scan + 1회 집계 | hit=28 | 1.016 ms | 12개 |

**결과 해석:** 조건 충족 상품은 12개이며 1위 Product 491은 평점 4.85·리뷰 160개이고 77.7% 단축되었다.

# 11. Q9 - 쿠폰 사용 여부별 AOV

## 요청사항

쿠폰 사용 주문과 미사용 주문의 주문당 평균 금액을 비교한다.

Q9 튜닝 전

```sql
-- ==========================================================
-- Q9. 쿠폰 사용 여부별 평균 주문금액
-- 문제점: 상세 행 조인 후 COUNT(DISTINCT)로 주문 수를 복원한다.
-- 의미: AOV는 전체 매출을 중복 없는 주문 수로 나눈 주문당 평균금액이다.
-- ==========================================================
-- [① 실행계획 확인용] 상세행 조인과 DISTINCT 집계 비용 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)                    -- 상세행 조인과 중복 제거 비용 측정
SELECT CASE WHEN o.coupon_code IS NULL THEN '미사용' ELSE '사용' END AS coupon_group, -- 쿠폰 여부 분류
       COUNT(DISTINCT o.order_id) AS order_count,    -- 중복을 제거한 주문 수
       ROUND(SUM(oi.line_total) / NULLIF(COUNT(DISTINCT o.order_id), 0), 2) AS aov -- 주문당 평균금액
FROM ecom.orders o
JOIN ecom.order_items oi ON oi.order_id = o.order_id
WHERE o.order_status IN ('paid', 'shipped', 'delivered')
GROUP BY 1
ORDER BY 1;
```

```sql
SELECT CASE WHEN o.coupon_code IS NULL THEN '미사용' ELSE '사용' END AS coupon_group, -- [②] 실제 쿠폰 그룹 비교 결과
       COUNT(DISTINCT o.order_id) AS order_count,
       ROUND(SUM(oi.line_total) / NULLIF(COUNT(DISTINCT o.order_id), 0), 2) AS aov
FROM ecom.orders o
JOIN ecom.order_items oi ON oi.order_id = o.order_id
WHERE o.order_status IN ('paid', 'shipped', 'delivered')
GROUP BY 1
ORDER BY 1;
```

orders(+) 1

SELECT CASE WHEN o.coupon_code IS NULL THEN '미사용' ELSE '사용' EN | Enter a SQL expression to filter results (use Ctrl+Space)

| | coupon_group | order_count | aov |
|---|---|---|---|
| 1 | 미사용 | 2,307 | 619.55 |
| 2 | 사용 | 761 | 1,694.79 |

Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) SELECT CASE WHEN o.coupo

| | QUERY PLAN |
|---|---|
| 1 | GroupAggregate  (cost=898.29..980.71 rows=1 width=72) (actual time=17.358..17.942 rows=2 loops=1) |
| 2 | Group Key: (CASE WHEN (o.coupon_code IS NULL) THEN '미사용'::text ELSE '사용'::text END) |
| 3 | Buffers: shared hit=143 |
| 4 | -> Sort  (cost=898.29..918.89 rows=8240 width=46) (actual time=16.192..16.553 rows=8230 loops=1) |
| 5 | Sort Key: (CASE WHEN (o.coupon_code IS NULL) THEN '미사용'::text ELSE '사용'::text END), o.order_id |
| 6 | Sort Method: quicksort  Memory: 750kB |
| 7 | Buffers: shared hit=143 |
| 8 | -> Hash Join  (cost=131.43..362.34 rows=8240 width=46) (actual time=1.732..9.532 rows=8230 loops=1) |
| 9 | Hash Cond: (oi.order_id = o.order_id) |
| 10 | Buffers: shared hit=140 |
| 11 | -> Seq Scan on order_items oi  (cost=0.00..203.67 rows=10367 width=14) (actual time=0.006..5.943 rows=10367 loops=1) |
| 12 | Buffers: shared hit=100 |
| 13 | -> Hash  (cost=93.08..93.08 rows=3068 width=15) (actual time=1.702..1.703 rows=3068 loops=1) |
| 14 | Buckets: 4096  Batches: 1  Memory Usage: 158kB |
| 15 | Buffers: shared hit=40 |
| 16 | -> Seq Scan on orders o  (cost=0.00..93.08 rows=3068 width=15) (actual time=0.012..1.193 rows=3068 loops=1) |
| 17 | Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[])) |
| 18 | Rows Removed by Filter: 792 |
| 19 | Buffers: shared hit=40 |
| 20 | Planning: |
| 21 | Buffers: shared hit=150 dirtied=1 |
| 22 | Planning Time: 4.284 ms |
| 23 | Execution Time: 17.993 ms |

## 느린 이유와 개선 방법

상세행 조인 후에는 주문 수 복원을 위해 `COUNT(DISTINCT)`가 필요하다. 튜닝 후에는 주문별 금액을 먼저 계산한 뒤 쿠폰 사용 여부로 그룹핑해 `AVG(order_total)`을 계산한다.

Q9 튜닝 후

```sql
-- Q9. 주문별 매출 선집계로 DISTINCT 제거
-- 튜닝: 주문별 금액을 먼저 한 행으로 만든 뒤 그룹별 AVG(order_total)을 계산한다.
-- [① 실행계획 확인용] 주문별 선집계 후 AVG 계산 계획 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)
WITH order_totals AS (
    SELECT o.order_id,
           (o.coupon_code IS NOT NULL) AS used_coupon, -- 쿠폰 사용 여부를 TRUE/FALSE로 변환
           SUM(oi.line_total) AS order_total        -- 주문별 총금액
    FROM ecom.orders o
    JOIN ecom.order_items oi ON oi.order_id = o.order_id
    WHERE o.order_status IN ('paid', 'shipped', 'delivered')
    GROUP BY o.order_id, o.coupon_code
)
SELECT CASE WHEN used_coupon THEN '사용' ELSE '미사용' END AS coupon_group,
       COUNT(order_id) AS order_count,
       ROUND(AVG(order_total), 2) AS aov            -- 주문별 금액의 평균
FROM order_totals
GROUP BY used_coupon
ORDER BY coupon_group;
```

```sql
WITH order_totals AS (                              -- [②] 실제 쿠폰 그룹 비교 결과 조회
    SELECT o.order_id, (o.coupon_code IS NOT NULL) AS used_coupon,
           SUM(oi.line_total) AS order_total
    FROM ecom.orders o
    JOIN ecom.order_items oi ON oi.order_id = o.order_id
    WHERE o.order_status IN ('paid', 'shipped', 'delivered')
    GROUP BY o.order_id, o.coupon_code
)
SELECT CASE WHEN used_coupon THEN '사용' ELSE '미사용' END AS coupon_group,
       COUNT(order_id) AS order_count, ROUND(AVG(order_total), 2) AS aov
FROM order_totals
GROUP BY used_coupon
ORDER BY coupon_group;
```

Results 1

WITH order_totals AS ( SELECT o.order_id, (o.coupon_code IS NOT NULL | Enter a SQL expression to filter results

| | coupon_group | order_count | aov |
|---|---|---|---|
| 1 | 미사용 | 2,307 | 619.55 |
| 2 | 사용 | 761 | 1,694.79 |

Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) WITH order_totals AS ( SELEC

| | QUERY PLAN |
|---|---|
| 1 | Sort  (cost=495.62..495.63 rows=2 width=73) (actual time=9.883..9.885 rows=2 loops=1) |
| 2 | Sort Key: (CASE WHEN ((o.coupon_code IS NOT NULL)) THEN '사용'::text ELSE '미사용'::text END) |
| 3 | Sort Method: quicksort  Memory: 25kB |
| 4 | Buffers: shared hit=140 |
| 5 | -> HashAggregate  (cost=495.58..495.61 rows=2 width=73) (actual time=9.788..9.791 rows=2 loops=1) |
| 6 | Group Key: (o.coupon_code IS NOT NULL) |
| 7 | Batches: 1  Memory Usage: 24kB |
| 8 | Buffers: shared hit=140 |
| 9 | -> HashAggregate  (cost=403.54..441.89 rows=3068 width=48) (actual time=8.547..9.389 rows=3068 loops=1) |
| 10 | Group Key: o.order_id |
| 11 | Batches: 1  Memory Usage: 1393kB |
| 12 | Buffers: shared hit=140 |
| 13 | -> Hash Join  (cost=131.43..362.34 rows=8240 width=21) (actual time=1.424..5.397 rows=8230 loops=1) |
| 14 | Hash Cond: (oi.order_id = o.order_id) |
| 15 | Buffers: shared hit=140 |
| 16 | -> Seq Scan on order_items oi  (cost=0.00..203.67 rows=10367 width=14) (actual time=0.014..1.794 rows=10367 loops=1) |
| 17 | Buffers: shared hit=100 |
| 18 | -> Hash  (cost=93.08..93.08 rows=3068 width=15) (actual time=1.349..1.350 rows=3068 loops=1) |
| 19 | Buckets: 4096  Batches: 1  Memory Usage: 158kB |
| 20 | Buffers: shared hit=40 |
| 21 | -> Seq Scan on orders o  (cost=0.00..93.08 rows=3068 width=15) (actual time=0.008..0.945 rows=3068 loops=1) |
| 22 | Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[])) |
| 23 | Rows Removed by Filter: 792 |
| 24 | Buffers: shared hit=40 |
| 25 | Planning: |
| 26 | Buffers: shared hit=22 dirtied=1 |
| 27 | Planning Time: 0.613 ms |
| 28 | Execution Time: 10.057 ms |

| 그룹 | 주문 수 | AOV | 해석 |
|---|---|---|---|
| 쿠폰 미사용 | 2,307 | 619.55 | 비교 기준 |
| 쿠폰 사용 | 761 | 1,694.79 | 미사용 대비 약 2.74배 |

| 구분 | Buffers | Execution Time |
|---|---|---|
| 튜닝 전 | hit=143 | 17.993 ms |
| 튜닝 후 | hit=140 | 10.057 ms |

> 쿠폰 사용과 높은 AOV의 상관관계만 확인할 수 있으며, 쿠폰의 인과효과를 의미하지는 않는다.

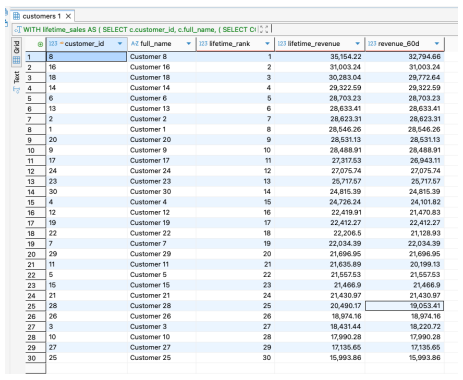
# 12. Q10 - 누적매출 상위 1% 고객의 최근 60일 매출

## 요청사항

전체 고객 3,000명을 누적매출로 정렬해 상위 1%인 정확히 30명을 먼저 선정하고, 선정된 고객의 최근 60일 매출을 제공한다.

Q10 튜닝 전

```sql
-- ============================================================
-- Q10. 누적매출 상위 1% 고객의 최근 60일 매출
-- 문제점: 3,000명 각각의 누적매출을 상관 서브쿼리로 계산하고,
--         선정된 상위 30명의 최근 60일 매출도 다시 상관 조회한다.
-- 기준: 전체 고객 3,000명의 누적매출로 정확히 30명을 먼저 선정한다.
-- 주의: 누적매출로 30명을 먼저 선정한 다음, 그 고객들의 최근 60일 매출을 표시한다.
-- ============================================================
-- [① 실행계획 확인용] 고객별 반복 집계와 최근 60일 SubPlan 비용 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)                  -- 고객별 반복 집계 비용 측정
WITH lifetime_sales AS (                             -- 고객별 누적매출 CTE
    SELECT c.customer_id,
           c.full_name,
           (
               SELECT COALESCE(SUM(oi.line_total), 0) -- 고객 한 명의 누적매출 계산
               FROM ecom.orders o
               JOIN ecom.order_items oi ON oi.order_id = o.order_id
               WHERE o.customer_id = c.customer_id
                 AND o.order_status IN ('paid', 'shipped', 'delivered')
           ) AS lifetime_revenue
    FROM ecom.customers c
), ranked AS (                                       -- 누적매출 순위 계산 CTE
    SELECT customer_id,
           full_name,
           lifetime_revenue,
           ROW_NUMBER() OVER (                       -- 중복 없는 순번을 1부터 부여
               ORDER BY lifetime_revenue DESC, customer_id
           ) AS lifetime_rank,
           COUNT(customer_id) OVER () AS customer_count -- 전체 고객 수를 각 행에 표시
    FROM lifetime_sales
)
SELECT r.customer_id,
       r.full_name,
       r.lifetime_rank,
       ROUND(r.lifetime_revenue, 2) AS lifetime_revenue,
       ROUND((
           SELECT COALESCE(SUM(oi.line_total), 0)
           FROM ecom.orders o
           JOIN ecom.order_items oi ON oi.order_id = o.order_id
           WHERE o.customer_id = r.customer_id
             AND o.order_status IN ('paid', 'shipped', 'delivered')
             AND o.order_ts >= now() - INTERVAL '60 days' -- 선정 고객의 최근 60일 매출
       ), 2) AS revenue_60d
FROM ranked r
WHERE r.lifetime_rank <= CEIL(r.customer_count * 0.01) -- 전체 고객의 상위 1%, 정확히 30명
ORDER BY r.lifetime_rank;
```



Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) WITH lifetime_sales AS ( SELE

```
Sort  (cost=167358.09..167360.59 rows=1000 width=93) (actual time=22.813..22.819 rows=30 loops=1)
  Sort Key: r.lifetime_rank
  Sort Method: quicksort  Memory: 27kB
  Buffers: shared hit=21777
  ->  Subquery Scan on r  (cost=100456.54..167308.26 rows=1000 width=93) (actual time=19.459..22.799 rows=30 loops=1)
        Filter: ((r.lifetime_rank)::numeric <= ceil(((r.customer_count)::numeric * 0.01)))
        Rows Removed by Filter: 2970
        Buffers: shared hit=21777
        ->  WindowAgg  (cost=100456.54..150539.51 rows=3000 width=69) (actual time=19.321..19.811 rows=3000 loops=1)
              Buffers: shared hit=19085
              ->  WindowAgg  (cost=50309.53..100435.76 rows=3000 width=61) (actual time=17.614..18.614 rows=3000 loops=1)
                    Buffers: shared hit=19085
                    ->  Sort  (cost=50309.51..50317.01 rows=3000 width=53) (actual time=17.607..17.779 rows=3000 loops=1)
                          Sort Key: ((SubPlan 2)) DESC, c.customer_id
                          Sort Method: quicksort  Memory: 235kB
                          Buffers: shared hit=19085
                          ->  Seq Scan on customers c  (cost=0.00..50136.25 rows=3000 width=53) (actual time=0.395..16.115 rows=30
                                Buffers: shared hit=19085
                                SubPlan 2
                                  ->  Aggregate  (cost=16.68..16.69 rows=1 width=32) (actual time=0.004..0.004 rows=1 loops=3000)
                                        Buffers: shared hit=19045
                                        ->  Nested Loop  (cost=0.57..16.67 rows=3 width=6) (actual time=0.002..0.004 rows=3 loops=3000)
                                              Buffers: shared hit=19045
                                              ->  Index Scan using idx_orders_customer_ts on orders o_1  (cost=0.28..8.30 rows=1 width=8) (act
                                                    Index Cond: (customer_id = c.customer_id)
                                                    Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                                                    Rows Removed by Filter: 0
                                                    Buffers: shared hit=9787
                                              ->  Index Scan using idx_order_items_order on order_items oi_1  (cost=0.29..8.34 rows=3 width=1
                                                    Index Cond: (order_id = o_1.order_id)
                                                    Buffers: shared hit=9258
```

```
      SubPlan 1
        ->  Aggregate  (cost=16.69..16.70 rows=1 width=32) (actual time=0.070..0.071 rows=1 loops=30)
              Buffers: shared hit=2692
              ->  Nested Loop  (cost=0.57..16.68 rows=3 width=6) (actual time=0.006..0.063 rows=63 loops=30)
                    Buffers: shared hit=2692
                    ->  Index Scan using idx_orders_customer_ts on orders o  (cost=0.28..8.31 rows=1 width=8) (actual time=0.004..
                          Index Cond: ((customer_id = r.customer_id) AND (order_ts >= (now() - '60 days'::interval)))
                          Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                          Rows Removed by Filter: 0
                          Buffers: shared hit=663
                    ->  Index Scan using idx_order_items_order on order_items oi  (cost=0.29..8.34 rows=3 width=14) (actual time=0
                          Index Cond: (order_id = o.order_id)
                          Buffers: shared hit=2029
Planning:
  Buffers: shared hit=140
Planning Time: 3.815 ms
Execution Time: 23.321 ms
```

## 느린 이유와 개선 방법

튜닝 전에는 고객 3,000명의 누적매출과 상위 고객의 최근 매출을 상관 서브쿼리로 반복 계산한다. 튜닝 후에는 전체 고객을 기준으로 누적매출과 최근 60일 매출을 한 번에 집계하고, `ROW_NUMBER()`와 전체 고객 수를 이용해 30명을 정확히 추출한다.

Q10 튜닝 후

```
-- Q10. 전체 3,000명의 누적매출과 최근 60일 매출을 한 번에 집계한 뒤
--      누적매출 상위 1%(30명)를 정확히 선정
-- 튜닝: 누적매출과 60일 매출을 한 번의 GROUP BY로 동시에 계산한다.
-- [① 실행계획 확인용] 고객별 SubPlan 없이 한 번에 집계하는지 확인
EXPLAIN (ANALYZE, BUFFERS, SUMMARY)
WITH customer_sales AS (
    SELECT c.customer_id,
           c.full_name,
           COALESCE(SUM(oi.line_total), 0) AS lifetime_revenue, -- 고객 누적매출
           COALESCE(SUM(oi.line_total) FILTER (
               WHERE o.order_ts >= now() - INTERVAL '60 days'
           ), 0) AS revenue_60d                    -- 같은 집계에서 최근 60일 매출만 합산
    FROM ecom.customers c
    LEFT JOIN ecom.orders o                        -- 주문 없는 고객도 전체 3,000명에 포함
      ON o.customer_id = c.customer_id
     AND o.order_status IN ('paid', 'shipped', 'delivered')
    LEFT JOIN ecom.order_items oi ON oi.order_id = o.order_id
    GROUP BY c.customer_id, c.full_name
), ranked AS (
    SELECT customer_id, full_name, lifetime_revenue, revenue_60d,
           ROW_NUMBER() OVER (                     -- 누적매출 순서대로 중복 없는 순번 부여
               ORDER BY lifetime_revenue DESC, customer_id
           ) AS lifetime_rank,
           COUNT(customer_id) OVER () AS customer_count -- 전체 고객 수를 각 행에 표시
    FROM customer_sales
)
SELECT customer_id, full_name, lifetime_rank,
       ROUND(lifetime_revenue, 2) AS lifetime_revenue,
       ROUND(revenue_60d, 2) AS revenue_60d
FROM ranked
WHERE lifetime_rank <= CEIL(customer_count * 0.01) -- 전체 고객 중 상위 1%, 정확히 30명
ORDER BY lifetime_rank;
```

```
WITH customer_sales AS (                          -- [②] 실제 상위 30명 결과 조회
    SELECT c.customer_id, c.full_name,
           COALESCE(SUM(oi.line_total), 0) AS lifetime_revenue,
           COALESCE(SUM(oi.line_total) FILTER (
               WHERE o.order_ts >= now() - INTERVAL '60 days'
           ), 0) AS revenue_60d
    FROM ecom.customers c
    LEFT JOIN ecom.orders o
      ON o.customer_id = c.customer_id
     AND o.order_status IN ('paid', 'shipped', 'delivered')
    LEFT JOIN ecom.order_items oi ON oi.order_id = o.order_id
    GROUP BY c.customer_id, c.full_name
), ranked AS (
    SELECT customer_id, full_name, lifetime_revenue, revenue_60d,
           ROW_NUMBER() OVER (
               ORDER BY lifetime_revenue DESC, customer_id
           ) AS lifetime_rank,
           COUNT(customer_id) OVER () AS customer_count
    FROM customer_sales
)
SELECT customer_id, full_name, lifetime_rank,
       ROUND(lifetime_revenue, 2) AS lifetime_revenue,
       ROUND(revenue_60d, 2) AS revenue_60d
FROM ranked
WHERE lifetime_rank <= CEIL(customer_count * 0.01)
ORDER BY lifetime_rank;
```

customers 1

WITH customer_sales AS ( SELECT c.customer_id, c.full_name, COALESC

| | customer_id | full_name | lifetime_rank | lifetime_revenue | revenue_60d |
|---|---|---|---|---|---|
| 1 | 8 | Customer 8 | 1 | 35,154.22 | 32,794.66 |
| 2 | 16 | Customer 16 | 2 | 31,003.24 | 31,003.24 |
| 3 | 18 | Customer 18 | 3 | 30,283.04 | 29,772.64 |
| 4 | 14 | Customer 14 | 4 | 29,322.59 | 29,322.59 |
| 5 | 6 | Customer 6 | 5 | 28,703.23 | 28,703.23 |
| 6 | 13 | Customer 13 | 6 | 28,633.41 | 28,633.41 |
| 7 | 2 | Customer 2 | 7 | 28,623.31 | 28,623.31 |
| 8 | 1 | Customer 1 | 8 | 28,546.26 | 28,546.26 |
| 9 | 20 | Customer 20 | 9 | 28,531.13 | 28,531.13 |
| 10 | 9 | Customer 9 | 10 | 28,488.91 | 28,488.91 |
| 11 | 17 | Customer 17 | 11 | 27,317.53 | 26,943.11 |
| 12 | 24 | Customer 24 | 12 | 27,075.74 | 27,075.74 |
| 13 | 23 | Customer 23 | 13 | 25,717.57 | 25,717.57 |
| 14 | 30 | Customer 30 | 14 | 24,815.39 | 24,815.39 |
| 15 | 4 | Customer 4 | 15 | 24,726.24 | 24,101.82 |
| 16 | 12 | Customer 12 | 16 | 22,419.91 | 21,470.83 |
| 17 | 19 | Customer 19 | 17 | 22,412.27 | 22,412.27 |
| 18 | 22 | Customer 22 | 18 | 22,206.5 | 21,128.93 |
| 19 | 7 | Customer 7 | 19 | 22,034.39 | 22,034.39 |
| 20 | 29 | Customer 29 | 20 | 21,696.95 | 21,696.95 |
| 21 | 11 | Customer 11 | 21 | 21,635.89 | 20,199.13 |
| 22 | 5 | Customer 5 | 22 | 21,557.53 | 21,557.53 |
| 23 | 15 | Customer 15 | 23 | 21,466.9 | 21,466.9 |
| 24 | 21 | Customer 21 | 24 | 21,430.97 | 21,430.97 |
| 25 | 28 | Customer 28 | 25 | 20,490.17 | 19,053.41 |
| 26 | 26 | Customer 26 | 26 | 18,974.16 | 18,974.16 |
| 27 | 3 | Customer 3 | 27 | 18,431.44 | 18,220.72 |
| 28 | 10 | Customer 10 | 28 | 17,990.28 | 17,990.28 |
| 29 | 27 | Customer 27 | 29 | 17,135.65 | 17,135.65 |
| 30 | 25 | Customer 25 | 30 | 15,993.86 | 15,993.86 |

Results 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) WITH customer_sales AS ( SE

```
QUERY PLAN
Sort  (cost=1053.20..1055.70 rows=1000 width=93) (actual time=16.928..16.932 rows=30 loops=1)
  Sort Key: ranked.lifetime_rank
  Sort Method: quicksort  Memory: 27kB
  Buffers: shared hit=180
  ->  Subquery Scan on ranked  (cost=930.83..1003.37 rows=1000 width=93) (actual time=15.860..16.914 rows=30 loops=1)
        Filter: ((ranked.lifetime_rank)::numeric <= ceil(((ranked.customer_count)::numeric * 0.01)))
        Rows Removed by Filter: 2970
        Buffers: shared hit=180
        ->  WindowAgg  (cost=930.83..930.87 rows=3000 width=101) (actual time=15.857..16.256 rows=3000 loops=1)
              Buffers: shared hit=180
              ->  WindowAgg  (cost=833.39..893.37 rows=3000 width=93) (actual time=14.683..15.357 rows=3000 loops=1)
                    Buffers: shared hit=180
                    ->  Sort  (cost=833.37..840.87 rows=3000 width=85) (actual time=14.679..14.788 rows=3000 loops=1)
                          Sort Key: customer_sales.lifetime_revenue DESC, customer_sales.customer_id
                          Sort Method: quicksort  Memory: 246kB
                          Buffers: shared hit=180
                          ->  Subquery Scan on customer_sales  (cost=615.11..660.11 rows=3000 width=85) (actual time=12.711..13.595 rows=3000 loops=1)
                                Buffers: shared hit=180
                                ->  HashAggregate  (cost=615.11..660.11 rows=3000 width=85) (actual time=12.710..13.399 rows=3000 loops=1)
                                      Group Key: c.customer_id
                                      Batches: 1  Memory Usage: 1905kB
                                      Buffers: shared hit=180
                                      ->  Hash Right Join  (cost=238.93..491.51 rows=8240 width=35) (actual time=3.716..8.317 rows=9013 loops=1)
                                            Hash Cond: (o.customer_id = c.customer_id)
                                            Buffers: shared hit=180
                                            ->  Hash Right Join  (cost=131.43..362.34 rows=8240 width=22) (actual time=1.959..4.969 rows=8230 loops=1)
                                                  Hash Cond: (oi.order_id = o.order_id)
                                                  Buffers: shared hit=140
                                                  ->  Seq Scan on order_items oi  (cost=0.00..203.67 rows=10367 width=14) (actual time=0.021..1.052 rows=10367 loop
                                                        Buffers: shared hit=100
                                                  ->  Hash  (cost=93.08..93.08 rows=3068 width=24) (actual time=1.912..1.912 rows=3068 loops=1)
                                                        Buckets: 4096  Batches: 1  Memory Usage: 200kB
                                                        Buffers: shared hit=40
                                                        ->  Seq Scan on orders o  (cost=0.00..93.08 rows=3068 width=24) (actual time=0.008..1.308 rows=3068 loops=1)
                                                              Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                                                              Rows Removed by Filter: 792
                                                              Buffers: shared hit=40
                                            ->  Hash  (cost=70.00..70.00 rows=3000 width=21) (actual time=1.717..1.718 rows=3000 loops=1)
                                                  Buckets: 4096  Batches: 1  Memory Usage: 190kB
                                                  Buffers: shared hit=40
                                                  ->  Seq Scan on customers c  (cost=0.00..70.00 rows=3000 width=21) (actual time=0.028..0.929 rows=3000 loops=1)
                                                        Buffers: shared hit=40
Planning:
  Buffers: shared hit=33
Planning Time: 1.384 ms
Execution Time: 17.230 ms
```

| 검증 항목 | 기대값 | 실제값 |
|---|---|---|
| 전체 고객 수 | 3,000 | 3,000 |
| 상위 1% 고객 수 | 30 | 30 |
| 누적매출 1위 고객 | - | Customer 8 / 35,154.22 |
| 해당 고객 최근 60일 매출 | - | 32,794.66 |

| 구분 | 반복 SubPlan | Buffers | Execution Time |
|---|---|---|---|
| 튜닝 전 | 있음(3,000회) | hit=21,777 | 23.321 ms |
| 튜닝 후 | 없음 | hit=180 | 17.230 ms |

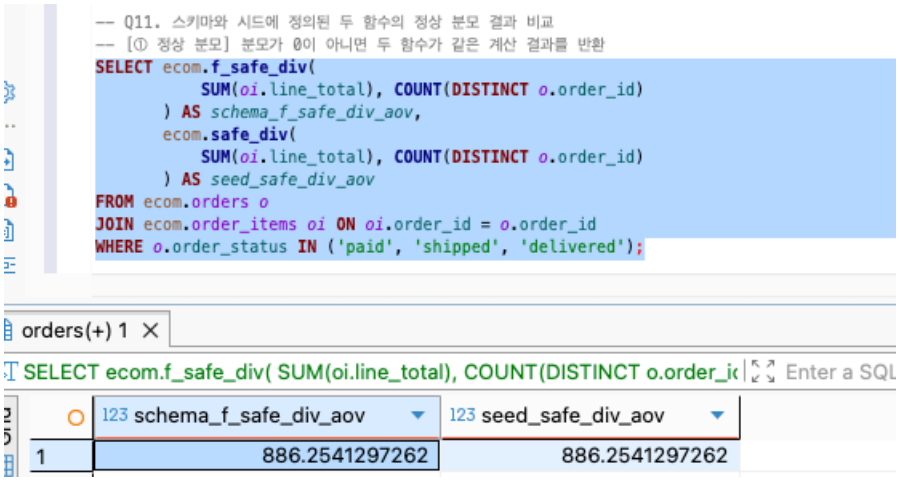
**결과 해석:** 전체 3,000명에서 누적매출 상위 30명을 정확히 선정했고 실행시간은 26.1% 단축되었다.

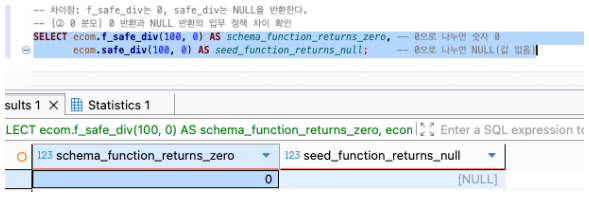
# 13. Q11 - 안전 나눗셈 함수 비교

## 함수 정의 차이

| 함수 | 정의 파일 | 분모가 0일 때 |
|---|---|---|
| `ecom.f_safe_div` | 스키마 파일 | `0` 반환 |
| `ecom.safe_div` | 시드 파일 | `NULL` 반환 |

Q11 함수 비교




## 실행 결과

| 테스트 | `f_safe_div` | `safe_div` | 해석 |
|---|---|---|---|
| 정상 AOV | 886.2541297262 | 886.2541297262 | 분모가 0이 아니면 같은 계산 결과 |
| `100 / 0` | `0` | `NULL` | 보고서 요구에 맞는 정책을 선택해야 함 |

# 14. 물리 조인 3종 비교

## 실행 결과 화면

Nested Loop Join

Hash Join

```sql
-- ==========================================================
-- 2. Hash Join
-- ==========================================================
-- 처리 방식: 작은 입력의 조인 키로 메모리에 해시 테이블을 만들고 큰 입력의 키를 대조한다.
-- 유리한 조건: 정렬되지 않은 대량 데이터의 동등 조인이고 해시 테이블이 메모리에 들어갈 때 유리하다.
-- 확인할 실행계획: Hash Join, Hash, Hash Cond

BEGIN;                                          -- Hash Join 테스트용 독립 트랜잭션 시작
SET LOCAL enable_hashjoin = on;                 -- Hash Join 후보 활성화
SET LOCAL enable_mergejoin = off;               -- Merge Join 후보 비활성화
SET LOCAL enable_nestloop = off;                -- Nested Loop Join 후보 비활성화

EXPLAIN (ANALYZE, BUFFERS, SETTINGS, SUMMARY)   -- 실제 Hash Join 계획과 실행 통계 출력
SELECT ROUND(SUM(oi.line_total), 2) AS total_sales_30d -- 세 조인에서 동일해야 하는 최근 30일 총매출
FROM ecom.orders o                              -- 필터 조건을 가진 주문 테이블
JOIN ecom.order_items oi                        -- 주문에 속한 상품별 금액 테이블
  ON oi.order_id = o.order_id                   -- Hash Cond로 사용할 주문번호 동등 조건
WHERE o.order_status IN ('paid', 'shipped', 'delivered') -- 취소·환불·생성 상태를 매출에서 제외
  AND o.order_ts >= CURRENT_DATE - 30;          -- 최근 30일 범위만 남겨 조인 대상 축소

ROLLBACK;                                       -- Hash Join 강제 설정 해제
```

Results 1 | Statistics 1

BEGIN; SET LOCAL enable_hashjoin = on; SET LOCAL enable_mergejoin = | Enter a SQL expression to filter results (use Ctrl+Space)

```
QUERY PLAN
Aggregate  (cost=292.06..292.08 rows=1 width=32) (actual time=11.540..11.541 rows=1 loops=1)
  Buffers: shared hit=109
  ->  Hash Join  (cost=55.14..286.05 rows=2404 width=6) (actual time=0.369..11.260 rows=2538 loops=1)
        Hash Cond: (oi.order_id = o.order_id)
        Buffers: shared hit=109
        ->  Seq Scan on order_items oi  (cost=0.00..203.67 rows=10367 width=14) (actual time=0.041..9.457 rows=10367 loops=1)
              Buffers: shared hit=100
        ->  Hash  (cost=43.95..43.95 rows=895 width=8) (actual time=0.316..0.316 rows=937 loops=1)
              Buckets: 1024  Batches: 1  Memory Usage: 45kB
              Buffers: shared hit=9
              ->  Index Only Scan using idx_orders_sales_ts_order on orders o  (cost=0.28..43.95 rows=895 width=8) (actual time=0.090..0.217 rows=937 loops=1)
                    Index Cond: (order_ts >= (CURRENT_DATE - 30))
                    Heap Fetches: 0
                    Buffers: shared hit=9
Settings: search_path = 'ecom, public', enable_mergejoin = 'off', enable_nestloop = 'off'
Planning Time: 0.294 ms
Execution Time: 11.590 ms
```

Merge Join

```sql
-- ==========================================================
-- 3. Merge Join
-- ==========================================================
-- 처리 방식: 양쪽 입력을 조인 키 순서로 정렬한 뒤 앞에서부터 순차적으로 비교한다.
-- 유리한 조건: 입력이 이미 조인 키로 정렬돼 있거나 정렬 비용이 낮은 대량 조인에 유리하다.
-- 확인할 실행계획: Merge Join, Merge Cond, 필요하면 Sort 노드

BEGIN;                                          -- Merge Join 테스트용 독립 트랜잭션 시작
SET LOCAL enable_hashjoin = off;                -- Hash Join 후보 비활성화
SET LOCAL enable_mergejoin = on;                -- Merge Join 후보 활성화
SET LOCAL enable_nestloop = off;                -- Nested Loop Join 후보 비활성화

EXPLAIN (ANALYZE, BUFFERS, SETTINGS, SUMMARY)   -- 실제 Merge Join 계획과 실행 통계 출력
SELECT ROUND(SUM(oi.line_total), 2) AS total_sales_30d -- 동일한 총매출을 계산하여 결과의 일관성 확인
FROM ecom.orders o                              -- 주문 테이블
JOIN ecom.order_items oi                        -- 주문상품 테이블
  ON oi.order_id = o.order_id                   -- Merge Cond로 사용할 주문번호 동등 조건
WHERE o.order_status IN ('paid', 'shipped', 'delivered') -- 실매출 상태만 포함
  AND o.order_ts >= CURRENT_DATE - 30;          -- 최근 30일 주문만 포함

ROLLBACK;                                       -- Merge Join 강제 설정 해제
```

Results 1 | Statistics 1

BEGIN; SET LOCAL enable_hashjoin = off; SET LOCAL enable_mergejoin = | Enter a SQL expression to filter results (use Ctrl+Space)

```
QUERY PLAN
Aggregate  (cost=499.06..499.08 rows=1 width=32) (actual time=5.602..5.603 rows=1 loops=1)
  Buffers: shared hit=131
  ->  Merge Join  (cost=88.11..493.05 rows=2404 width=6) (actual time=0.305..5.346 rows=2538 loops=1)
        Merge Cond: (oi.order_id = o.order_id)
        Buffers: shared hit=131
        ->  Index Scan using idx_order_items_order on order_items oi  (cost=0.29..350.79 rows=10367 width=14) (actual time=0.030..4.060 rows=10367 loops=1)
              Buffers: shared hit=122
        ->  Sort  (cost=87.83..90.07 rows=895 width=8) (actual time=0.269..0.341 rows=937 loops=1)
              Sort Key: o.order_id
              Sort Method: quicksort  Memory: 25kB
              Buffers: shared hit=9
              ->  Index Only Scan using idx_orders_sales_ts_order on orders o  (cost=0.28..43.95 rows=895 width=8) (actual time=0.035..0.158 rows=937 loops=1)
                    Index Cond: (order_ts >= (CURRENT_DATE - 30))
                    Heap Fetches: 0
                    Buffers: shared hit=9
Settings: search_path = 'ecom, public', enable_hashjoin = 'off', enable_nestloop = 'off'
Planning:
  Buffers: shared hit=30
Planning Time: 1.964 ms
Execution Time: 5.705 ms
```

| 구분 | 처리 방식 | 일반적으로 유리한 조건 | Buffers | Execution Time |
|---|---|---|---|---|
| Nested Loop Join | 외부 행마다 내부 입력 탐색 | 외부 입력이 작고 내부 인덱스가 있을 때 | `hit=1,837/read=58` | `9.017` ms |
| Hash Join | 작은 입력 해시 생성 후 큰 입력 탐색 | 큰 비정렬 동등 조인 | `hit=109` | `11.590` ms |
| Merge Join | 정렬된 두 입력을 순차 병합 | 양쪽이 이미 정렬된 큰 입력 | `hit=131` | `5.705` ms |

## 본인 환경에서의 결론

동일 데이터에서는 Merge Join(5.705 ms)이 가장 빨랐고 Hash Join(11.590 ms)이 가장 느렸다. 단, 데이터 크기·정렬·인덱스에 따라 최적 조인은 달라진다.

# 15. Materialized View 활용

## 생성 목적

매일 주문과 주문상품을 다시 조인·집계하지 않고, 미리 계산한 일별 주문 수와 GMV를 빠르게 조회한다.

## 생성 및 결과 화면

MV 생성

Statistics 1

| Name | Value |
|---|---|
| Updated Rows | 0 |
| Execute time | 0.023s |
| Start time | Fri Aug 14 17:05:51 KST 2026 |
| Finish time | Fri Aug 14 17:05:51 KST 2026 |
| Query | DROP MATERIALIZED VIEW IF EXISTS ecom.mv_daily_gmv; -- 기존 MV와 그 MV에 속한 인덱스를 함께 제거 |
| | -- ========== |
| | -- 2. 일별 GMV Materialized View 생성 |
| | -- ========== |
| | -- WITH NO DATA로 구조와 정의만 먼저 생성하고 데이터는 뒤의 REFRESH에서 적재한다. |
| | CREATE MATERIALIZED VIEW ecom.mv_daily_gmv AS -- 일별 집계 결과를 저장할 MV 생성 시작 |
| | SELECT date_trunc('day', o.order_ts)::date AS sales_day, -- 주문시각을 일 단위로 자르고 date 형식으로 변환 |
| | COUNT(DISTINCT o.order_id) AS order_count, -- 주문상품 조인으로 중복된 주문번호를 제거한 일별 주문 수 |
| | ROUND(SUM(oi.line_total), 2) AS gmv -- 주문상품 금액의 일별 합계를 소수점 둘째 자리로 표시 |
| | FROM ecom.orders o -- 주문일시·주문상태가 저장된 주문 테이블 |
| | JOIN ecom.order_items oi -- 상품별 판매금액이 저장된 주문상품 테이블 |
| | ON oi.order_id = o.order_id -- 주문번호가 같은 주문과 주문상품을 연결 |
| | WHERE o.order_status IN ('paid', 'shipped', 'delivered') -- 실제 매출로 인정하는 상태만 포함 |
| | GROUP BY date_trunc('day', o.order_ts)::date -- 같은 날짜의 주문상품 금액을 한 행으로 집계 |
| | WITH NO DATA; -- 생성 시에는 데이터를 넣지 않고 빈 구조만 생성 |
| | -- ========== |
| | -- 3. 동시 갱신을 위한 UNIQUE 인덱스 생성 |
| | -- ========== |
| | -- REFRESH MATERIALIZED VIEW CONCURRENTLY를 사용하려면 모든 행을 유일하게 식별할 수 있어야 한다. |
| | -- 이 MV는 날짜별로 한 행만 존재하므로 sales_day를 고유키로 사용한다. |
| | CREATE UNIQUE INDEX ux_mv_daily_gmv_sales_day -- MV의 날짜 중복을 금지하는 UNIQUE 인덱스 이름 |
| | ON ecom.mv_daily_gmv (sales_day); -- 날짜 한 개가 MV의 각 행을 유일하게 식별 |
| | -- ========== |
| | -- 4. 최초 데이터 적재 및 통계정보 수집 |
| | -- ========== |
| | -- WITH NO DATA로 만들었으므로 최초 1회는 일반 REFRESH로 데이터를 채워야 한다. |
| | -- 빈 MV에는 CONCURRENTLY를 바로 사용할 수 없으므로 먼저 일반 REFRESH를 실행한다. |
| | REFRESH MATERIALIZED VIEW ecom.mv_daily_gmv; -- MV 정의 쿼리를 실행하여 일별 결과를 실제로 저장 |
| | ANALYZE ecom.mv_daily_gmv; -- MV의 행 수와 값 분포 통계를 수집하여 실행계획 선택에 활용 |

## 원본 쿼리 계획

```
-- ==========
-- 5. 원본 테이블을 직접 조인·집계하는 실행계획
-- ==========
-- 비교 기준이 되는 원본 쿼리이다.
-- orders와 order_items를 읽고 Join, Aggregate, Sort를 수행하므로 MV 조회보다 작업량이 많을 수 있다.
-- 캡처할 값: 최상위 Buffers, Planning Time, Execution Time, Join/Aggregate 노드

EXPLAIN (ANALYZE, BUFFERS, SUMMARY)                 -- 원본 쿼리를 실제 실행하여 버퍼와 실행시간 측정
SELECT date_trunc('day', o.order_ts)::date AS sales_day, -- 주문일을 일 단위로 변환
       COUNT(DISTINCT o.order_id) AS order_count,   -- 조인 후 중복을 제거한 일별 주문 수
       ROUND(SUM(oi.line_total), 2) AS gmv          -- 일별 총 상품 판매금액
FROM ecom.orders o                                  -- 원본 주문 테이블 읽기
JOIN ecom.order_items oi                            -- 원본 주문상품 테이블 읽기
  ON oi.order_id = o.order_id                       -- 주문번호를 기준으로 원본 테이블 두 개 조인
WHERE o.order_status IN ('paid', 'shipped', 'delivered') -- 실매출 상태만 집계
GROUP BY date_trunc('day', o.order_ts)::date        -- 날짜별로 주문 수와 GMV 계산
ORDER BY sales_day DESC                             -- 가장 최근 날짜부터 정렬
LIMIT 30;                                           -- 최근 30개 날짜만 출력
```

Results 1 | Statistics 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) SELECT date_trunc('day', o.or | Enter a SQL expression to filter results (use Ctrl+Space)

| | QUERY PLAN |
|---|---|
| 1 | Limit (cost=939.49..940.89 rows=30 width=44) (actual time=9.799..10.154 rows=30 loops=1) |
| 2 | Buffers: shared hit=140 |
| 3 | -> GroupAggregate (cost=939.49..1083.25 rows=3068 width=44) (actual time=9.797..10.149 rows=30 loops=1) |
| 4 | Group Key: ((date_trunc('day'::text, o.order_ts))::date) |
| 5 | Buffers: shared hit=140 |
| 6 | -> Sort (cost=939.49..960.09 rows=8240 width=18) (actual time=9.771..9.856 rows=1368 loops=1) |
| 7 | Sort Key: ((date_trunc('day'::text, o.order_ts))::date) DESC, o.order_id |
| 8 | Sort Method: quicksort Memory: 706kB |
| 9 | Buffers: shared hit=140 |
| 10 | -> Hash Join (cost=131.43..403.54 rows=8240 width=18) (actual time=2.372..7.234 rows=8230 loops=1) |
| 11 | Hash Cond: (oi.order_id = o.order_id) |
| 12 | Buffers: shared hit=140 |
| 13 | -> Seq Scan on order_items oi (cost=0.00..203.67 rows=10367 width=14) (actual time=0.034..1.046 rows=10367 loops=1) |
| 14 | Buffers: shared hit=100 |
| 15 | -> Hash (cost=93.08..93.08 rows=3068 width=16) (actual time=2.269..2.270 rows=3068 loops=1) |
| 16 | Buckets: 4096 Batches: 1 Memory Usage: 176kB |
| 17 | Buffers: shared hit=40 |
| 18 | -> Seq Scan on orders o (cost=0.00..93.08 rows=3068 width=16) (actual time=0.020..1.465 rows=3068 loops=1) |
| 19 | Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[])) |
| 20 | Rows Removed by Filter: 792 |
| 21 | Buffers: shared hit=40 |
| 22 | Planning: |
| 23 | Buffers: shared hit=18 |
| 24 | Planning Time: 0.835 ms |
| 25 | Execution Time: 10.257 ms |

```
-- ==========
-- 6. Materialized View 조회 실행계획
-- ==========
-- 이미 계산되어 저장된 sales_day, order_count, gmv만 읽는다.
-- 원본 테이블 조인과 GROUP BY가 사라지는지 원본 실행계획과 비교한다.
-- 캡처할 값: Scan 방식, 최상위 Buffers, Planning Time, Execution Time

EXPLAIN (ANALYZE, BUFFERS, SUMMARY)                 -- MV 조회를 실제 실행하여 버퍼와 실행시간 측정
SELECT sales_day,                                   -- MV에 저장된 매출 날짜
       order_count,                                 -- MV에 저장된 일별 주문 수
       gmv                                          -- MV에 저장된 일별 GMV
FROM ecom.mv_daily_gmv                              -- 원본 조인 대신 사전 집계된 MV만 조회
ORDER BY sales_day DESC                             -- 가장 최근 날짜부터 정렬
LIMIT 30;                                           -- 최근 30개 날짜만 출력
```

Results 1 | Statistics 1

EXPLAIN (ANALYZE, BUFFERS, SUMMARY) SELECT sales_day, order_coul | Enter a SQL expression to filter results (use Ctrl+Space)

| | QUERY PLAN |
|---|---|
| 1 | Limit (cost=0.14..3.26 rows=30 width=20) (actual time=0.014..0.018 rows=30 loops=1) |
| 2 | Buffers: shared hit=1 read=1 |
| 3 | -> Index Scan Backward using ux_mv_daily_gmv_sales_day on mv_daily_gmv (cost=0.14..14.17 rows=135 width=20) (actual time=0.013..0.016 rows=30 loops=1) |
| 4 | Buffers: shared hit=1 read=1 |
| 5 | Planning: |
| 6 | Buffers: shared hit=56 read=1 |
| 7 | Planning Time: 0.582 ms |
| 8 | Execution Time: 0.027 ms |

## MV 조회 결과

```
-- ==========
-- 7. Materialized View 실제 결과 확인
-- ==========
-- EXPLAIN 없이 실행하여 구매·재무팀에 전달할 실제 일별 결과를 확인한다.

SELECT sales_day,                                   -- 일별 매출 기준 날짜
       order_count,                                 -- 해당 날짜의 중복 없는 실매출 주문 수
       gmv                                          -- 해당 날짜의 주문상품 총매출
FROM ecom.mv_daily_gmv                              -- 미리 저장된 일별 집계 결과 조회
ORDER BY sales_day DESC                             -- 최근 날짜부터 표시
LIMIT 30;                                           -- 최근 30일분 결과 표시
```

mv_daily_gmv 1 | Statistics 1

SELECT sales_day, order_count, gmv FROM ecom.mv_daily_gmv ORDER E | Enter a SQL expression to filter results (use Ctrl+Space)

| | sales_day | order_count | gmv |
|---|---|---|---|
| 1 | 2026-09-04 | 1 | 234.98 |
| 2 | 2026-09-03 | 2 | 577.53 |
| 3 | 2026-08-30 | 2 | 492.73 |
| 4 | 2026-08-29 | 2 | 716.1 |
| 5 | 2026-08-28 | 1 | 339.12 |
| 6 | 2026-08-27 | 2 | 470.26 |
| 7 | 2026-08-26 | 2 | 493.92 |
| 8 | 2026-08-24 | 1 | 171.21 |
| 9 | 2026-08-22 | 3 | 400.5 |
| 10 | 2026-08-21 | 2 | 647.42 |
| 11 | 2026-08-20 | 6 | 1,434.01 |
| 12 | 2026-08-19 | 3 | 390.33 |
| 13 | 2026-08-18 | 2 | 294.15 |
| 14 | 2026-08-17 | 2 | 278.8 |
| 15 | 2026-08-14 | 21 | 16,900.68 |
| 16 | 2026-08-13 | 30 | 27,756.96 |
| 17 | 2026-08-12 | 22 | 18,433.21 |
| 18 | 2026-08-11 | 31 | 29,707.93 |
| 19 | 2026-08-10 | 25 | 16,932.28 |
| 20 | 2026-08-09 | 29 | 25,555.9 |
| 21 | 2026-08-08 | 30 | 27,412.01 |
| 22 | 2026-08-07 | 25 | 21,950.82 |
| 23 | 2026-08-06 | 42 | 39,935.38 |
| 24 | 2026-08-05 | 27 | 22,869.32 |
| 25 | 2026-08-04 | 27 | 32,913.44 |
| 26 | 2026-08-03 | 29 | 31,064.72 |
| 27 | 2026-08-02 | 35 | 29,385.72 |
| 28 | 2026-08-01 | 38 | 43,562.65 |
| 29 | 2026-07-31 | 40 | 43,292.45 |
| 30 | 2026-07-30 | 29 | 24,239.39 |

MV 갱신

```
-- ==========================================================
-- 8. 서비스 조회를 유지하면서 Materialized View 갱신
-- ==========================================================
-- CONCURRENTLY는 갱신 중에도 기존 MV에 대한 일반 조회를 가능하게 한다.
-- 이 명령을 사용하려면 위에서 생성한 UNIQUE 인덱스가 반드시 존재해야 한다.
-- 원본 데이터 변경분을 증분 반영하는 방식이 아니라 MV 전체 결과를 다시 계산한다.

REFRESH MATERIALIZED VIEW CONCURRENTLY ecom.mv_daily_gmv; -- 조회 잠금을 줄이면서 최신 원본 데이터로 갱신
```

Statistics 1

| Name | Value |
|---|---|
| Updated Rows | 0 |
| Execute time | 0.025s |
| Start time | Fri Aug 14 17:07:41 KST 2026 |
| Finish time | Fri Aug 14 17:07:41 KST 2026 |
| Query | REFRESH MATERIALIZED VIEW CONCURRENTLY ecom.mv_daily_gmv; -- 조회 잠금을 줄이면서 최신 원본 데이터로 갱신 |

| 구분 | 핵심 실행계획 | Buffers | Execution Time |
|---|---|---|---|
| 원본 JOIN + GROUP BY | Hash Join + GroupAgg | hit=140 | 10.257 ms |
| `mv_daily_gmv` 조회 | Index Scan Backward | hit=1/read=1 | 0.027 ms |

오후 3시 설계

```
-- ==========================================================
-- 9. 오후 3시 자동 갱신 설계
-- ==========================================================
-- pg_cron 확장 기능이 설치된 PostgreSQL 환경에서만 사용할 수 있다.
-- 로컬 환경에 pg_cron이 없다면 아래 코드는 주석 상태로 두고 실행하지 않는다.

-- CREATE EXTENSION IF NOT EXISTS pg_cron;          -- pg_cron 확장이 없으면 생성
-- SELECT cron.schedule(                             -- 정해진 시각에 실행할 작업 등록
--     'refresh-ecom-daily-gmv-1500',                -- 스케줄 작업을 식별하는 고유 이름
--     '0 15 * * *',                                 -- 매일 15시 00분에 실행하는 cron 표현식
--     $$REFRESH MATERIALIZED VIEW CONCURRENTLY ecom.mv_daily_gmv$$ -- 자동 실행할 갱신 명령
-- );

-- 등록 확인 예시
-- SELECT jobid, schedule, command
-- FROM cron.job;

-- 등록한 작업 해제 예시
-- SELECT cron.unschedule('refresh-ecom-daily-gmv-1500');
```

### 갱신 전략

- 최초 적재: `REFRESH MATERIALIZED VIEW ecom.mv_daily_gmv;`
- 운영 갱신: `REFRESH MATERIALIZED VIEW CONCURRENTLY ecom.mv_daily_gmv;`
- 제안 주기: 매일 오후 3시
- 주의점: 오후 3시 이후 변경사항은 다음 갱신 전까지 조회 결과에 반영되지 않는다.

## 16. DBMS별 옵티마이저 비교 (Option)

자세한 조사 내용은 `종합실습4_06_DBMS별_옵티마이저_비교.md`를 참고한다.

| DBMS | 대표 조인 | 실행계획 확인 | 특징 |
|---|---|---|---|
| PostgreSQL | Nested Loop, Hash, Merge | `EXPLAIN ANALYZE` | 통계·비용 기반, 확장 통계, GEQO |
| MySQL | Nested Loop/BKA, Hash | `EXPLAIN ANALYZE` | optimizer switch·힌트·히스토그램 |
| Oracle | Nested Loop, Hash, Sort Merge | `DBMS_XPLAN` | Adaptive Plan, SQL Plan Management |
| SQL Server | Nested Loops, Hash, Merge, Adaptive | Actual Execution Plan | Query Store, Intelligent Query Processing |

## 17. 전체 튜닝 전후 비교

| 문항 | 튜닝 전 방식 | 튜닝 후 방식 | 전 Buffers | 후 Buffers | 전 시간 | 후 시간 | 감소율 |
|---|---|---|---|---|---|---|---|
| Q1 | 함수 조건 + 상관 집계 | SARG 범위 + 단일 집계 | hit=2,902 | hit=101/read=8 | 25.380 | 4.903 | 80.7% |
| Q2 | 상세행 + DISTINCT | 주문별 선집계 | hit=140 | hit=143 | 11.646 | 6.879 | 40.9% |
| Q3 | 비-SARG 기간 조건 | 최근 주문 선필터 | hit=148 | hit=120/read=9 | 11.863 | 11.534 | 2.8% |
| Q4 | 제품별 상관 서브쿼리 | 전체 1회 집계 + RANK | hit=70,677 | hit=147 | 37.373 | 9.654 | 74.2% |
| Q5 | 고객별 3회 반복 조회 | 주문 선집계 + 고객 GROUP BY | hit=17,287 | hit=180 | 21.915 | 13.785 | 37.1% |
| Q6 | 고객별 EXISTS | Window + BOOL_OR | hit=9,362 | hit=1/read=20 | 7.881 | 6.166 | 21.8% |
| Q7 | 임계치 상관 조회 | 직접 비교 + 부분 인덱스 | hit=1,912 | hit=12 | 2.017 | 0.428 | 78.8% |
| Q8 | AVG/COUNT 반복 | GROUP BY + HAVING | hit=1,565 | hit=28 | 4.551 | 1.016 | 77.7% |
| Q9 | 상세행 + DISTINCT | 주문별 선집계 + AVG | hit=143 | hit=140 | 17.993 | 10.057 | 44.1% |
| Q10 | 누적/60일 매출 반복 | 전체 1회 집계 + 정확히 | hit=21,777 | hit=180 | 23.321 | 17.230 | 26.1% |

```
(튜닝 전 시간 - 튜닝 후 시간) / 튜닝 전 시간 × 100
```

# 18. 최종 분석

## 요청 대응 관점

어떤 주문 상태와 기간을 포함하는지, 주문 수와 주문상품 수를 혼동하지 않았는지, 상위 고객의 선정 기준과 표시 기간이 다른지 먼저 정의해야 한다. 특히 Q10은 최근 60일 매출로 상위 고객을 선정하는 것이 아니라 누적매출로 30명을 선정한 뒤 최근 60일 매출을 계산한다.

## 튜닝 관점

상관 SubPlan 제거와 선집계·SARG 조건 적용으로 Q1~Q10 모두 단축되었으며, 가장 큰 개선은 Q1(80.7%)이었다.

## 한계와 추가 확인사항

- 랜덤 시드이므로 재실행 시 결과가 달라진다.
- 로컬 단일 사용자 환경의 실행시간은 운영 부하를 반영하지 않는다.
- 쿠폰 사용 여부와 AOV의 관계는 인과관계가 아니다.
- 인덱스는 조회 성능을 높이지만 INSERT/UPDATE와 저장공간 비용을 추가한다.

## 최종 결론

업무 기준을 명시한 Q1~Q11 결과를 튜닝 전후 동일하게 유지하면서 반복 조회와 불필요한 중복 제거를 줄였다. 물리 조인은 본 환경에서 Merge Join이 가장 빨랐고, mv_daily_gmv 조회는 원본 집계 10.257 ms에서 0.027 ms로 단축되었다. 운영 시에는 데이터 규모와 변경 빈도에 따라 인덱스 비용과 오후 3시 MV 갱신 시차를 함께 관리해야 한다.